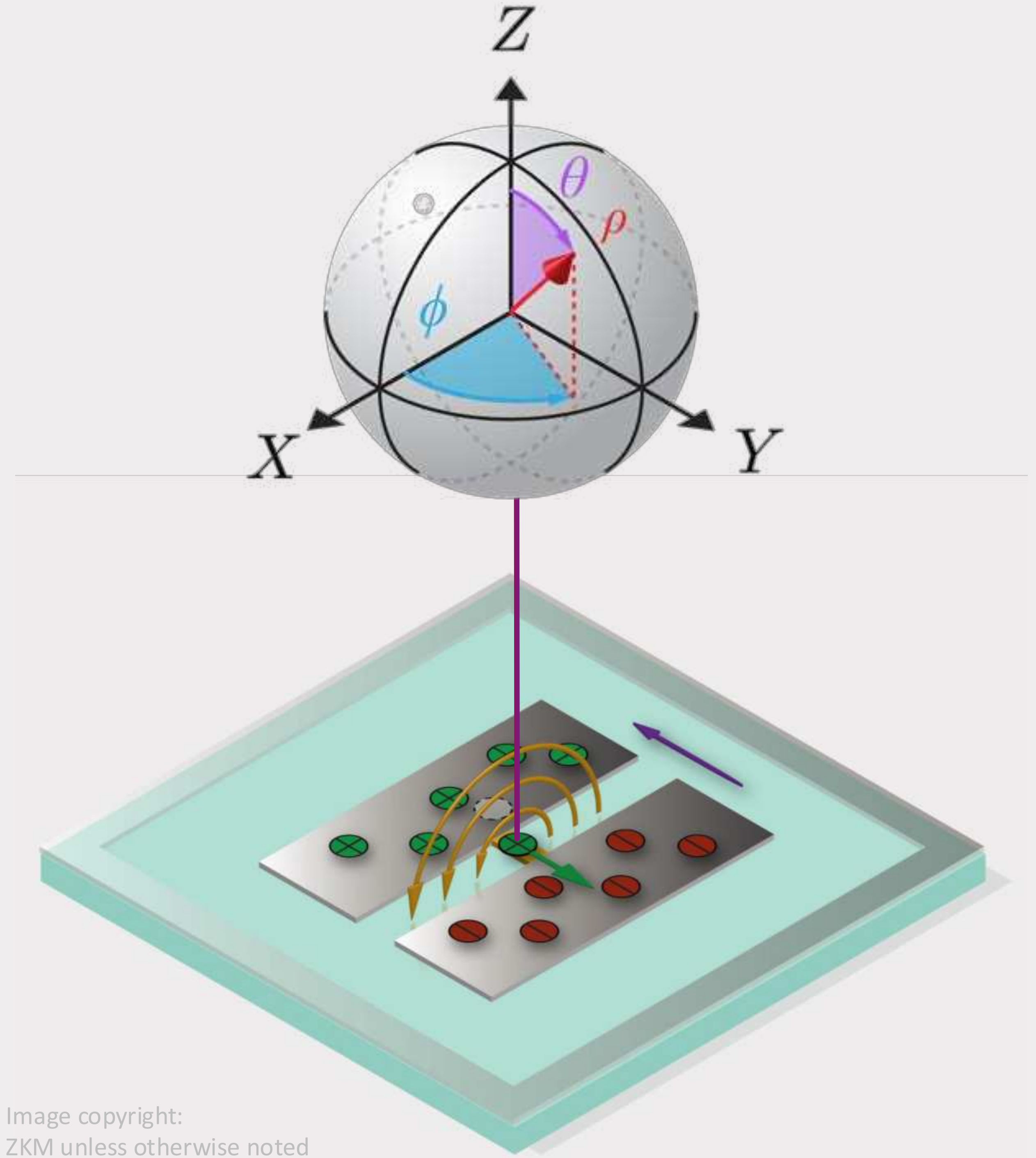


Your First Superconducting Qubit

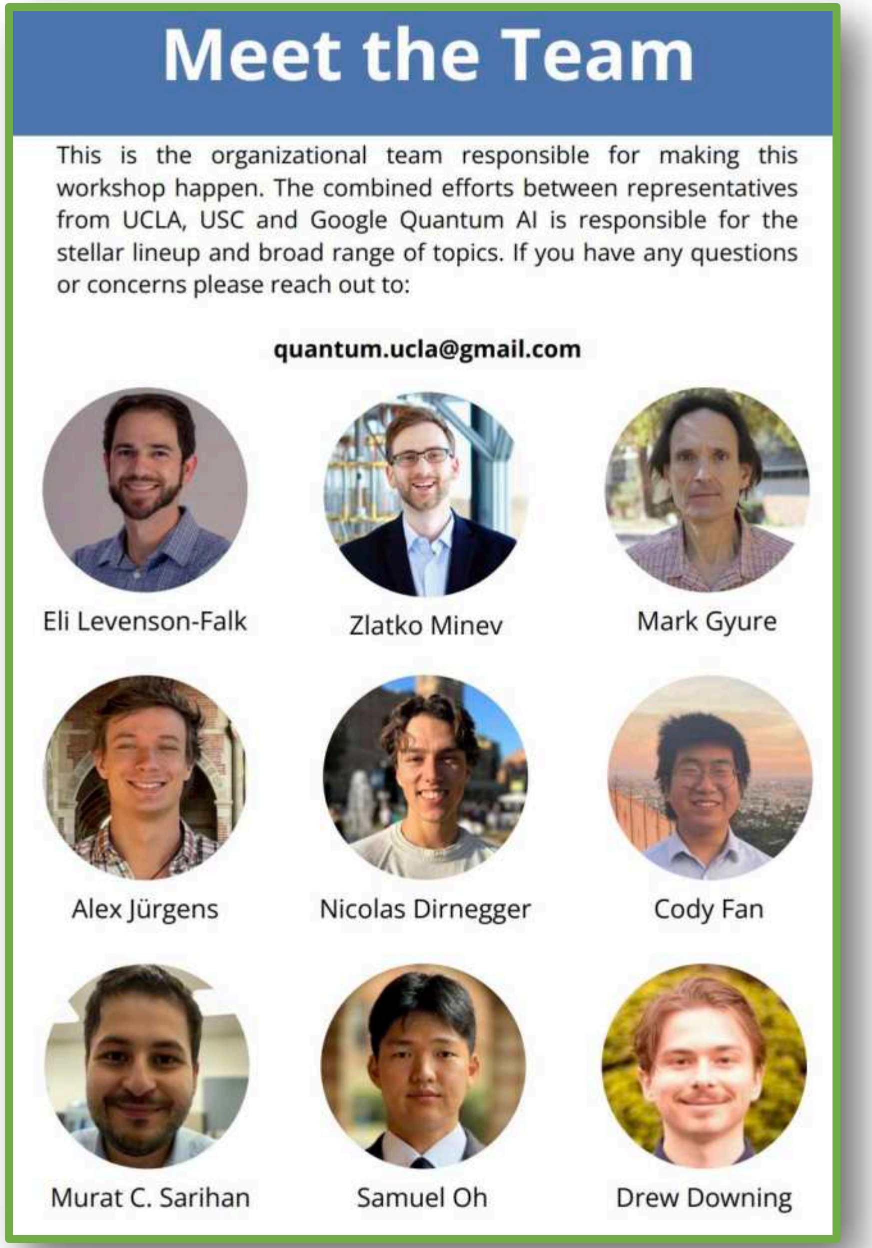
Introduction to **Circuit Quantum Electrodynamics (cQED)**

Zlatko K. Minev

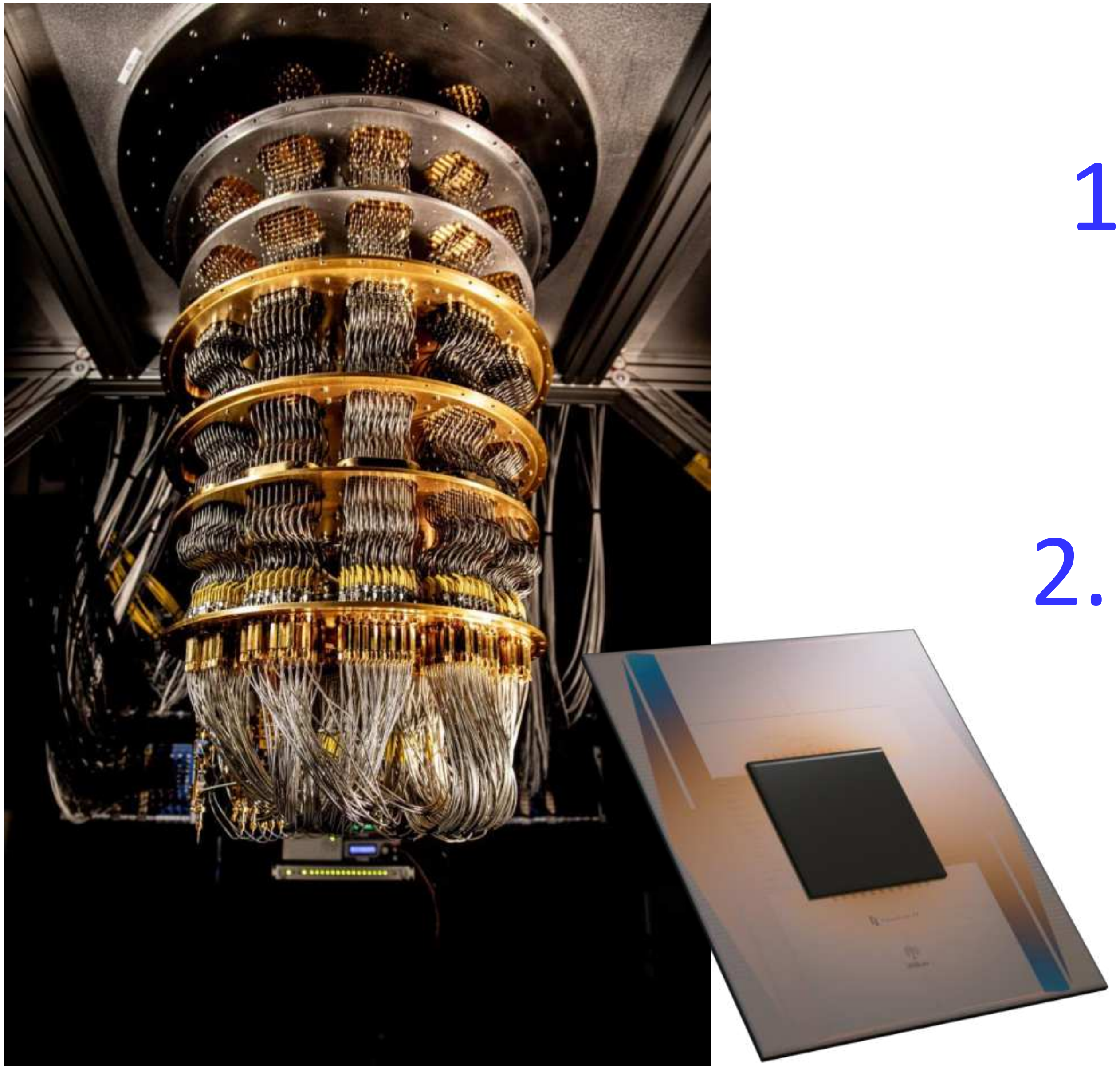
Current Affiliation:
Google Quantum AI; Assoc. Fellow @ CIFAR
Former Affiliation: IBM Quantum; Yale University

@zlatko_minev zlatko-minev.com/blog

Huge THANK YOU to the organizers and broader team!



Let's get to know each other ☺



Images: Google Quantum AI

1. Have you ever used a quantum device?
2. Analyzed/designed/made one?

This Lecture

Introductory and skill reaffirming

Don't need to know much going in, but we will go far

Advanced material

Examples: simplest, most practical examples

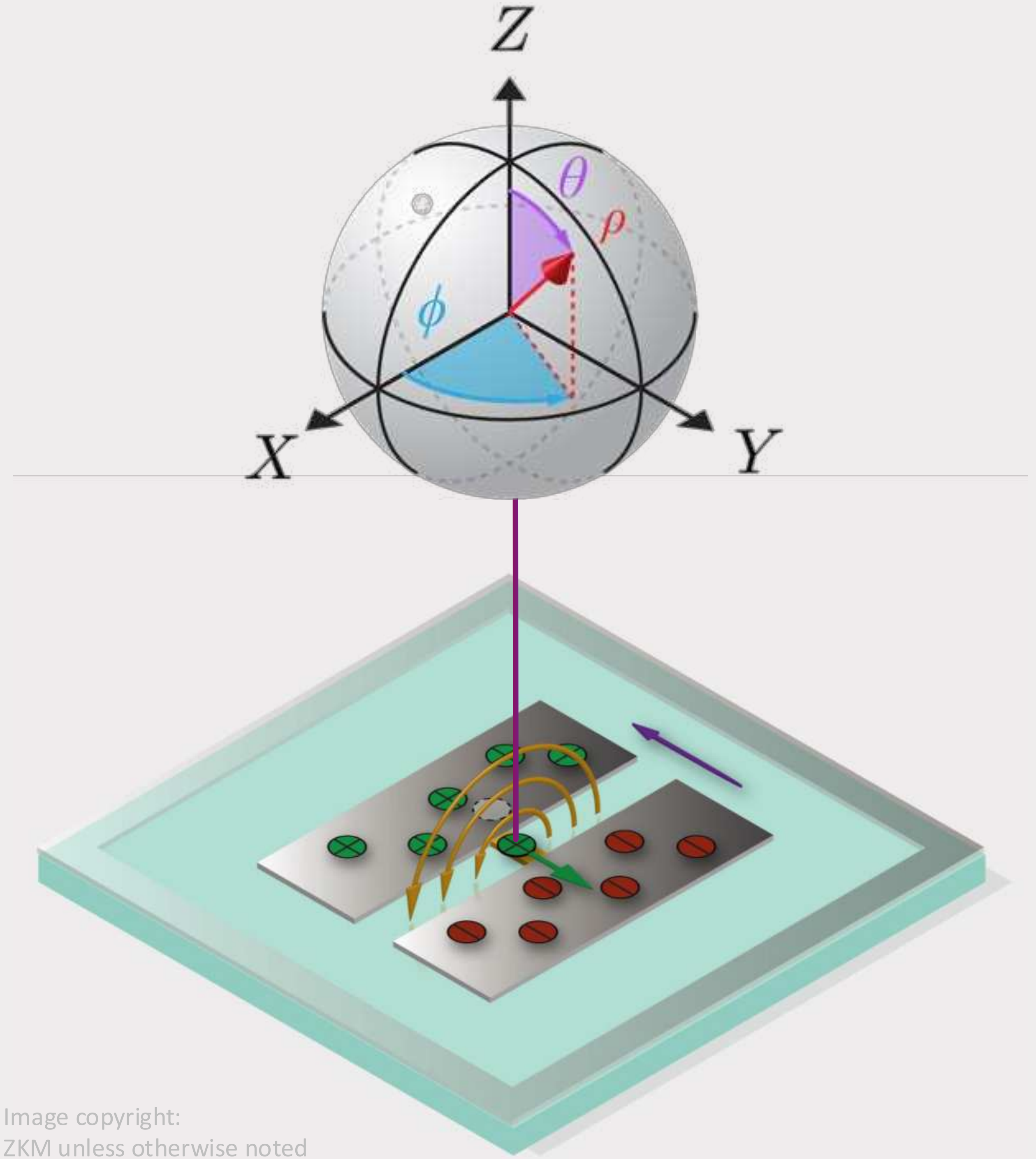
Step by step

Ask questions!

Lab work with Dr. Murat Sarıhan,
Sadman Shanto, and more!

Slides!

Icons here by FreePik. Dangerous bend symbol based on Donald Knuth.



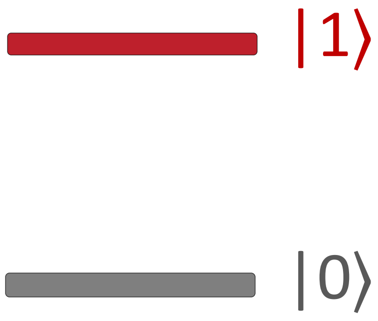
Qubit

From Idea to Reality

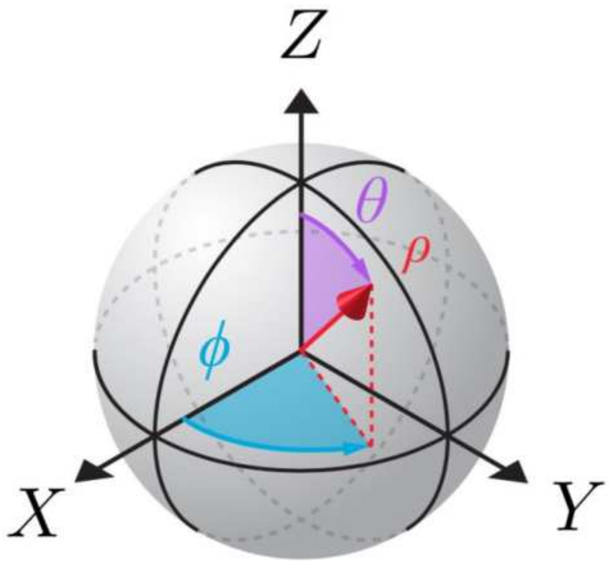
THE BIG PICTURE
before calculations

Qubit: idea

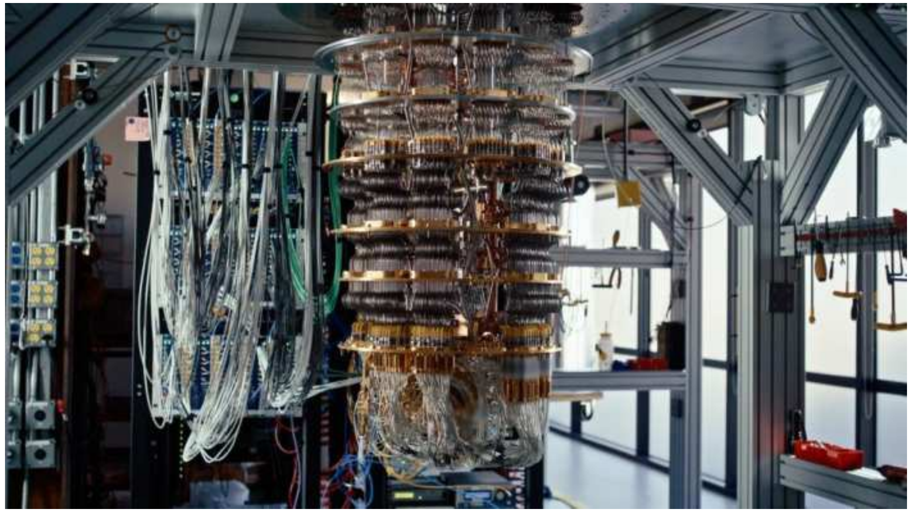
Energy levels



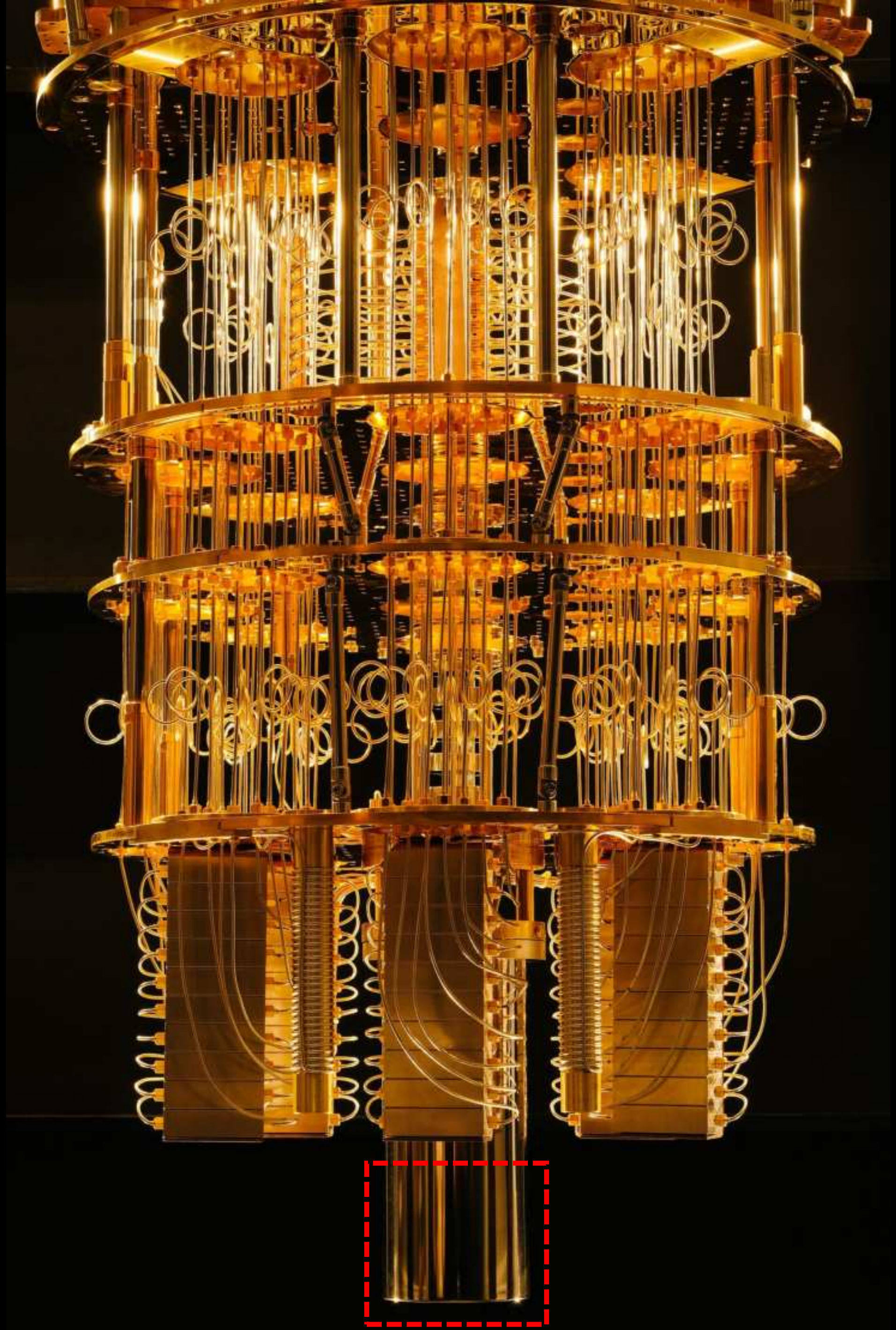
Hilbert space



Quantum cloud?



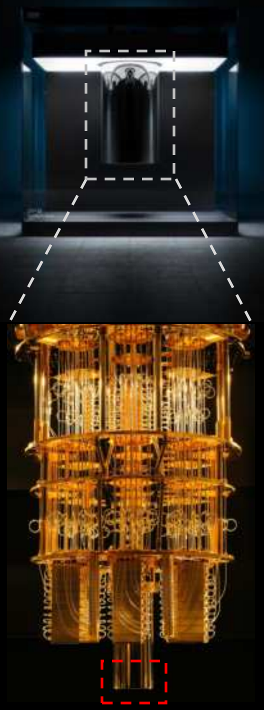
[*] Images: Left/Middle: Minev, arXiv:1902.10355; Right: Google Quantum AI

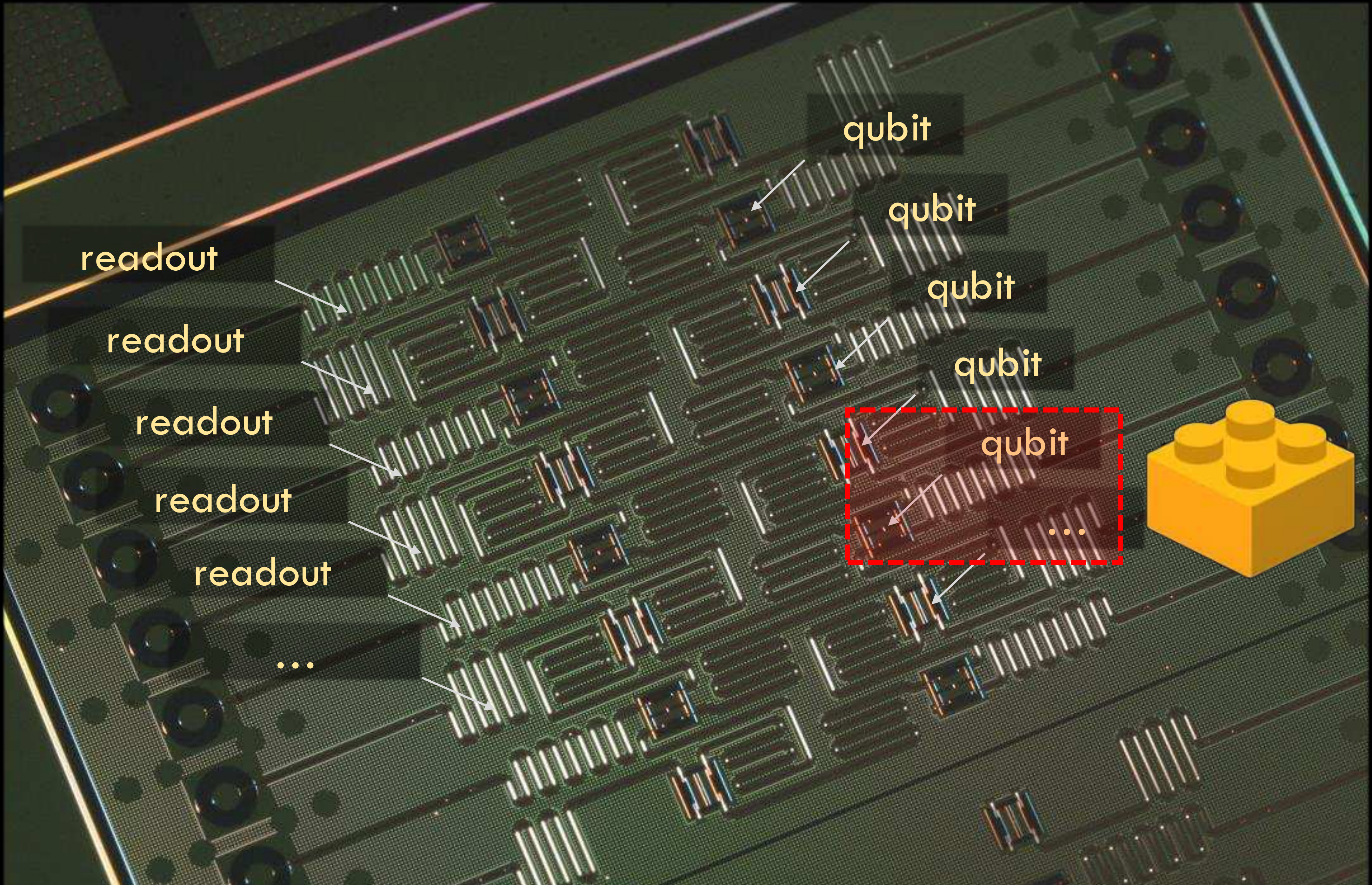


Operation at
15 mK (-273.13 °C)

Images: IBM

Zlatko Minev



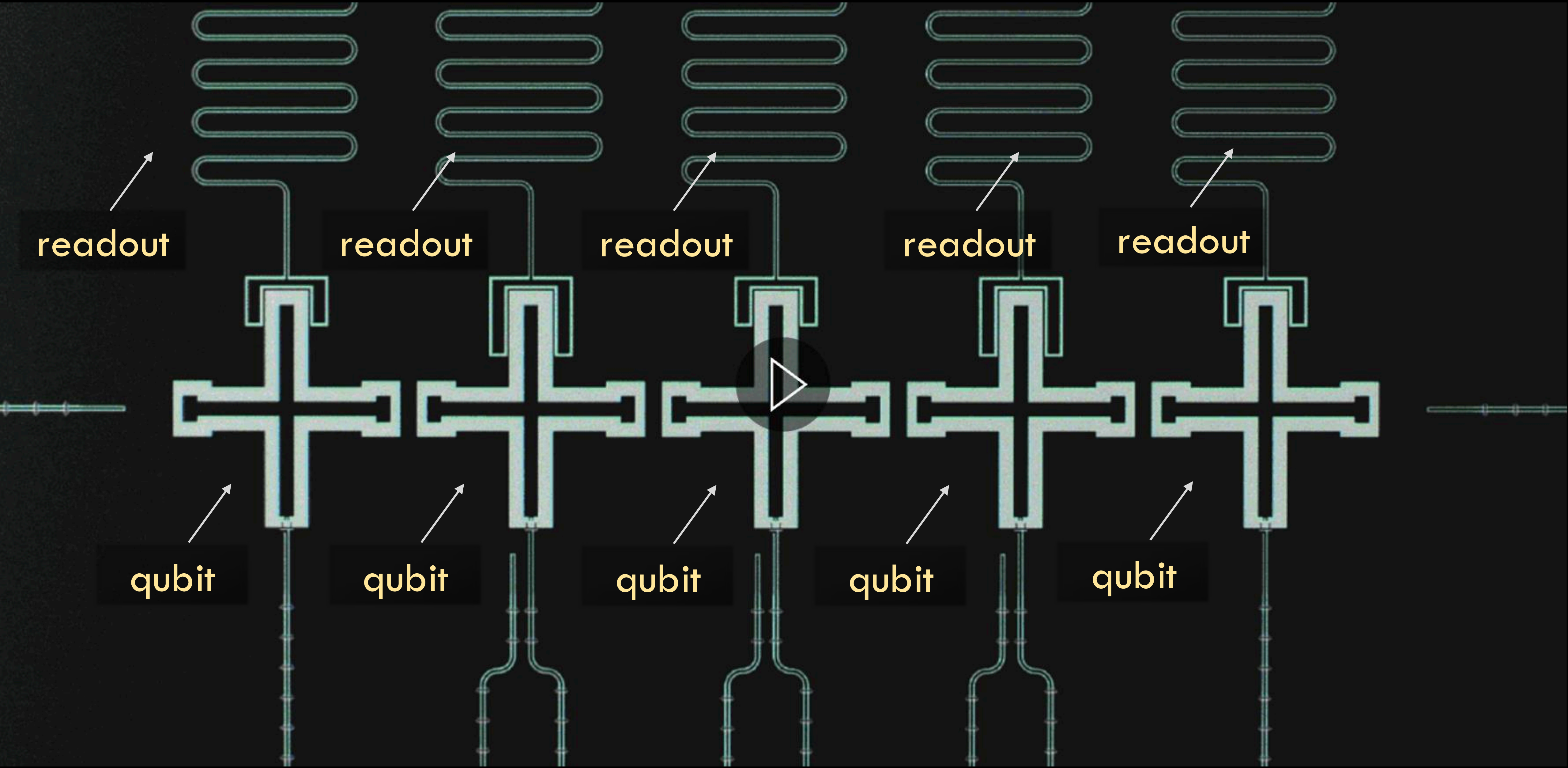


Example
IBM Architecture

Images: IBM

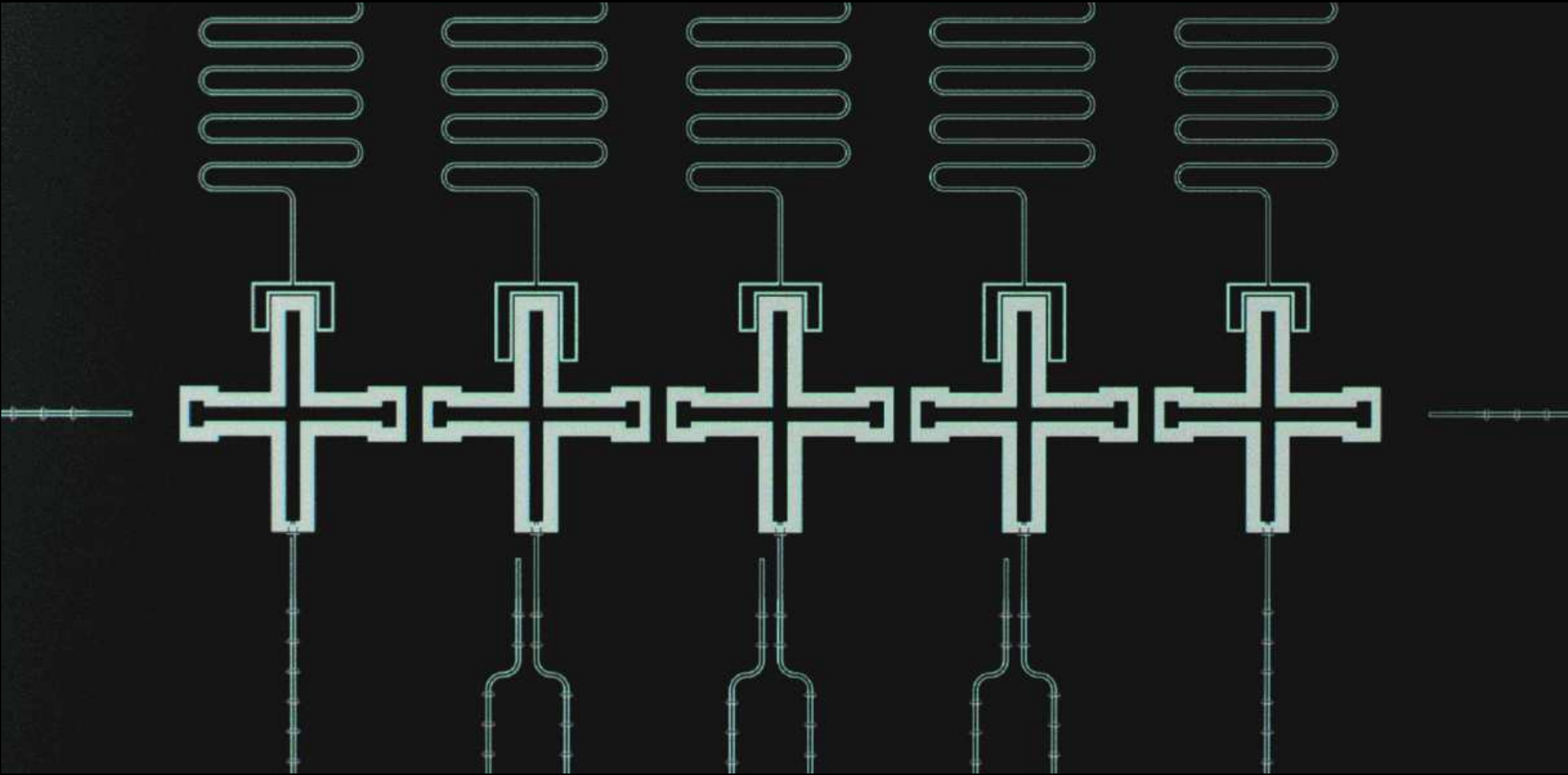
Zlatko Minev

Example Google Architecture



Video: Google

Example Google Architecture



Video: Google

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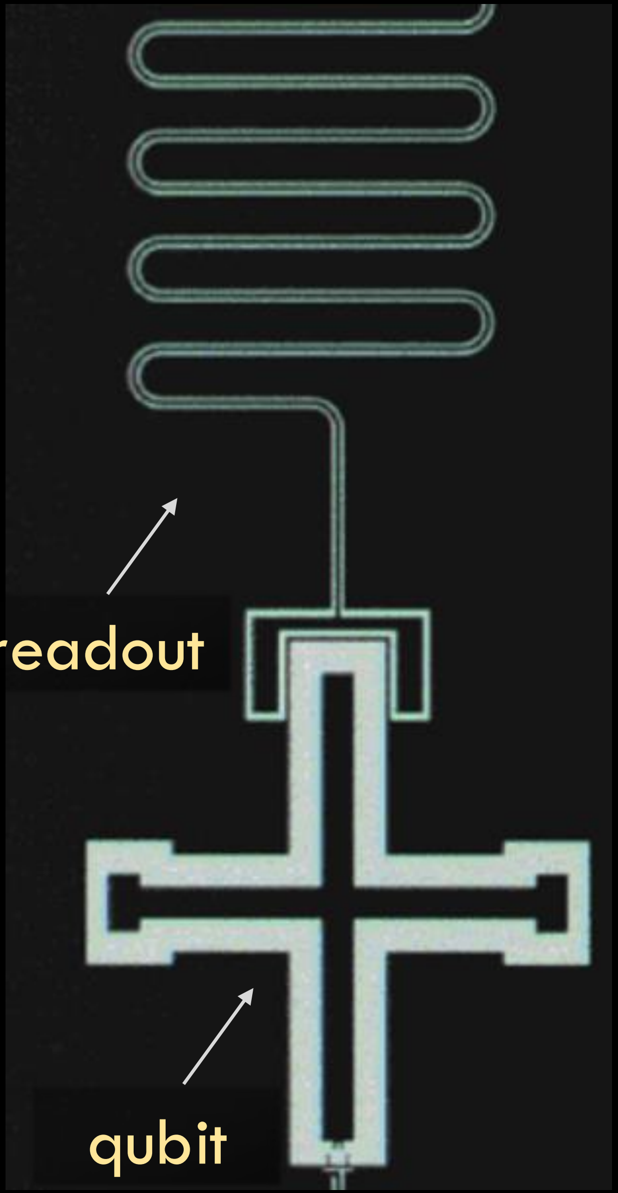
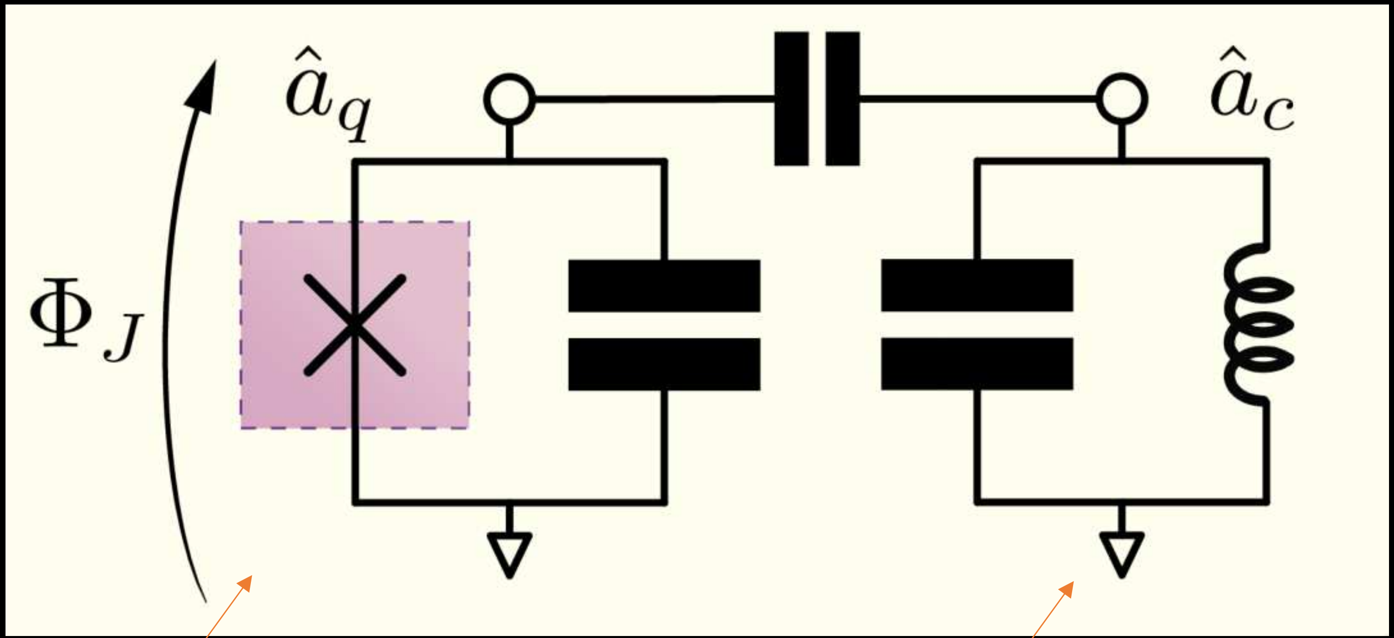
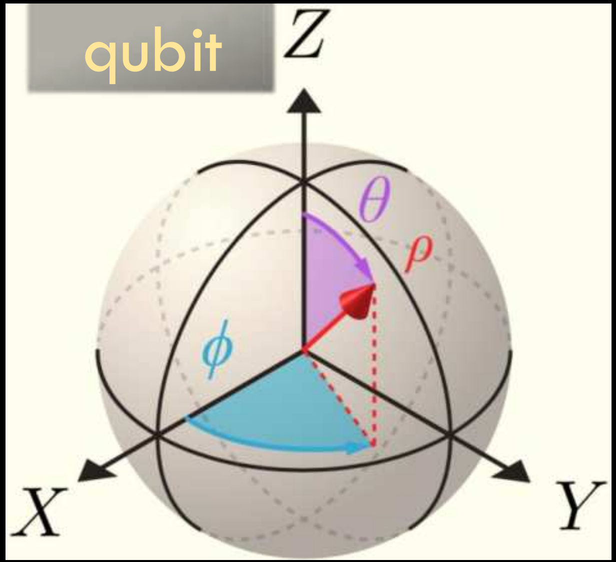
Physical and abstract circuits and Hilbert space

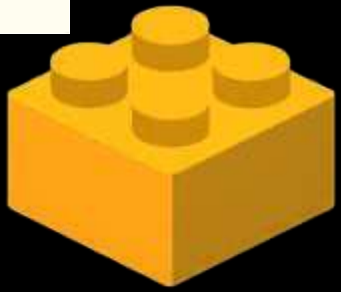




qubit



readout



$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

Video: Google

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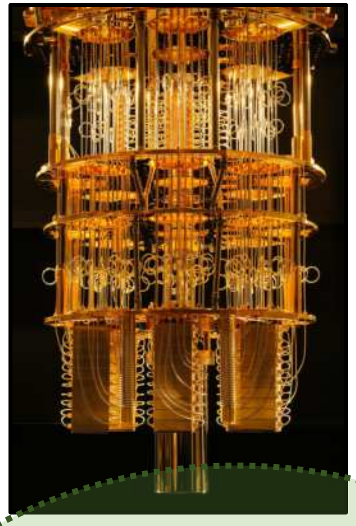
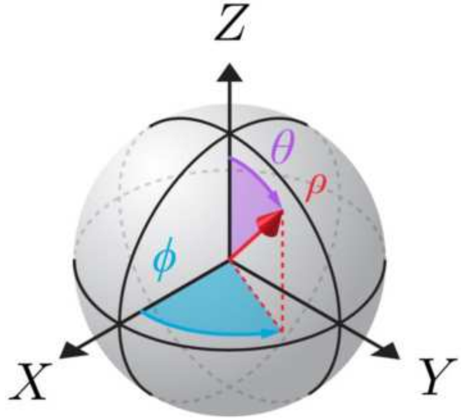
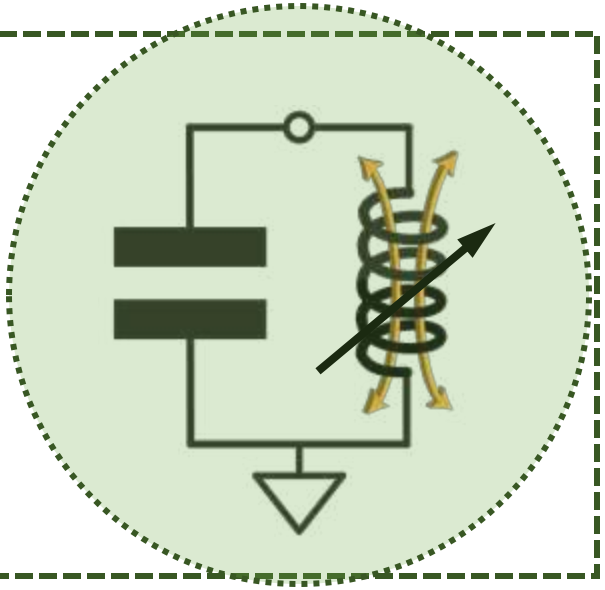
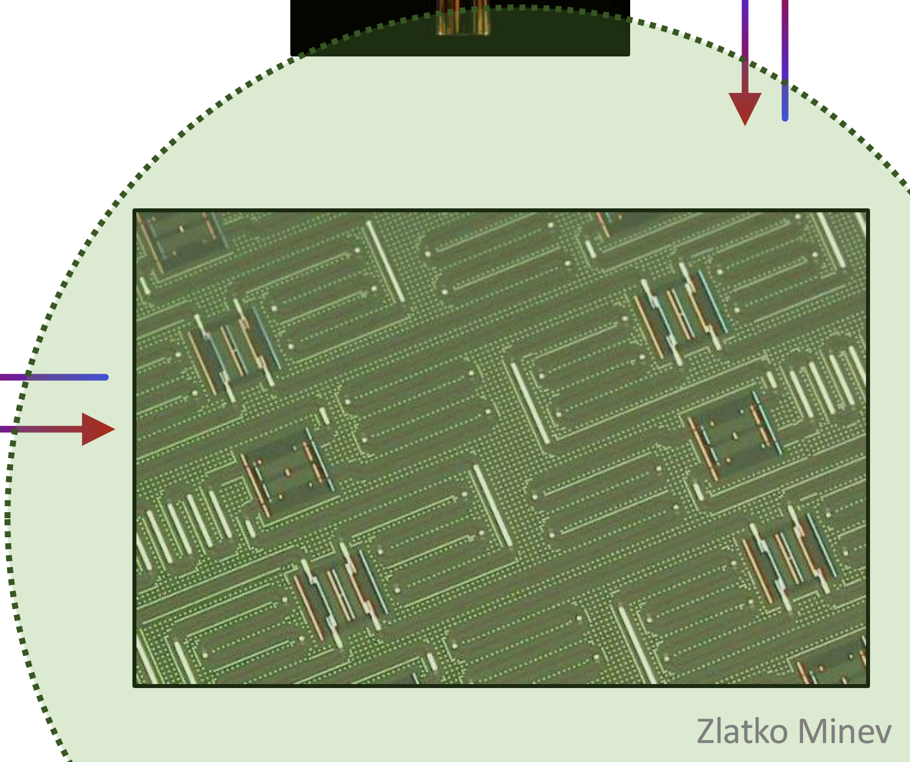
cQED qubit in the cloud: Summary of flow









* Laptop image: rawpixel.com; Photos: IBM

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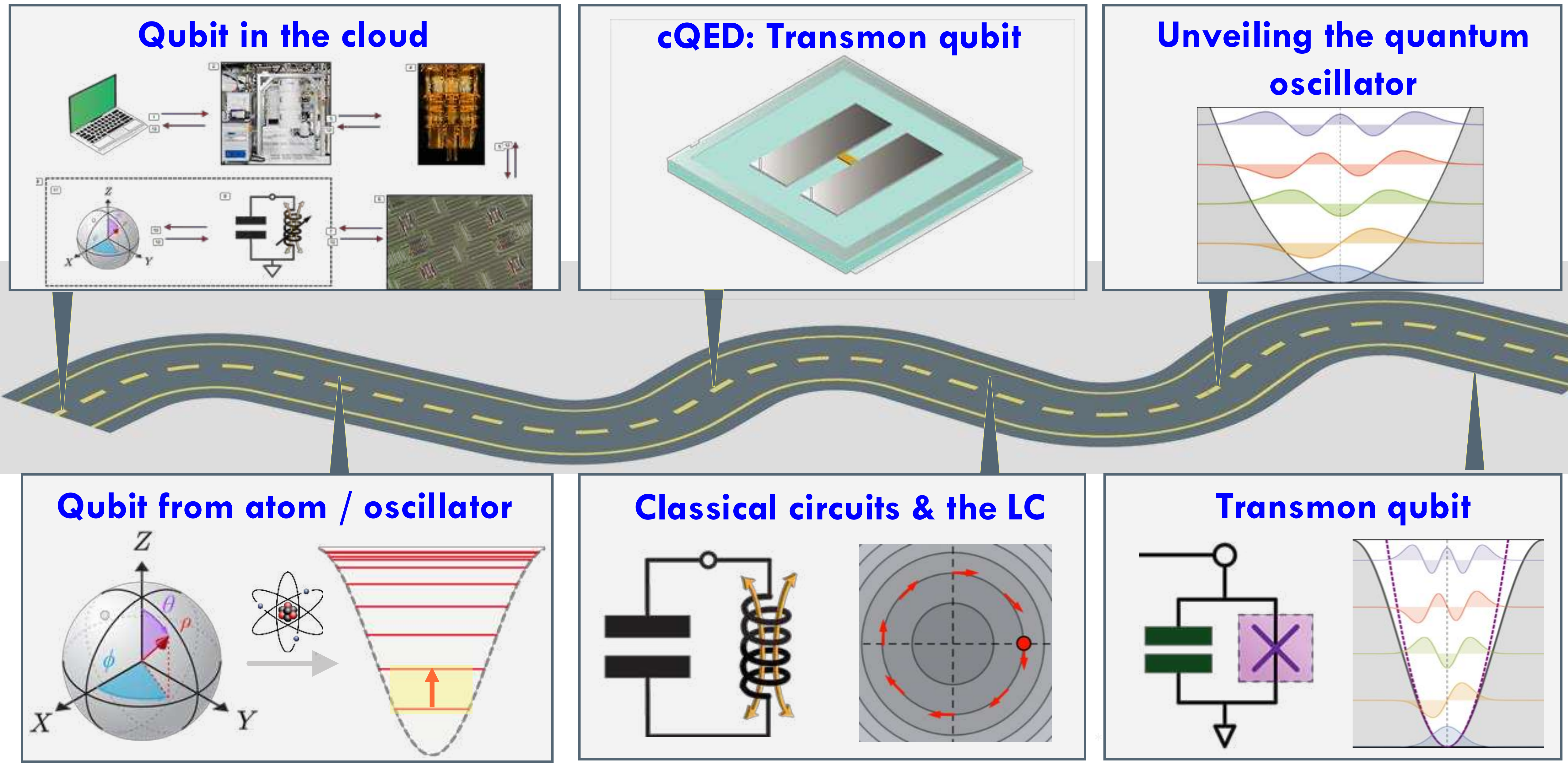
On the road ahead

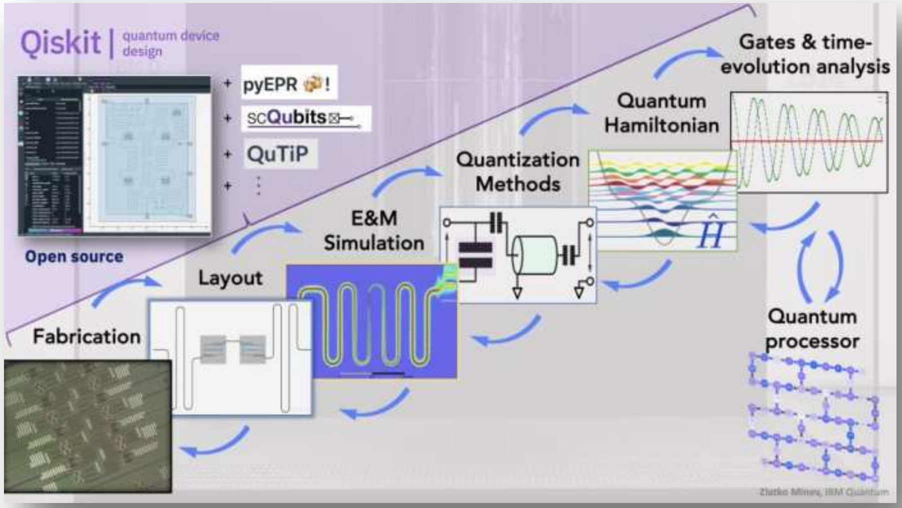


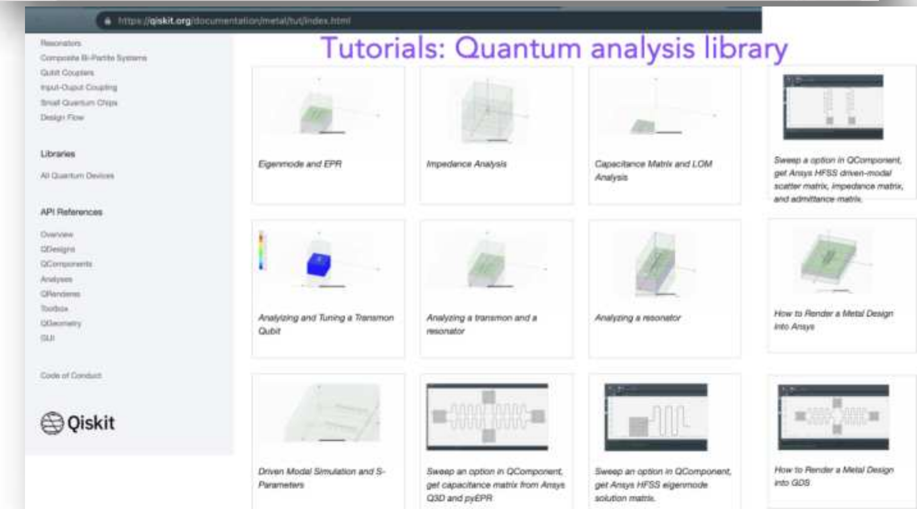
Automation in Practice: Qiskit Metal + Community

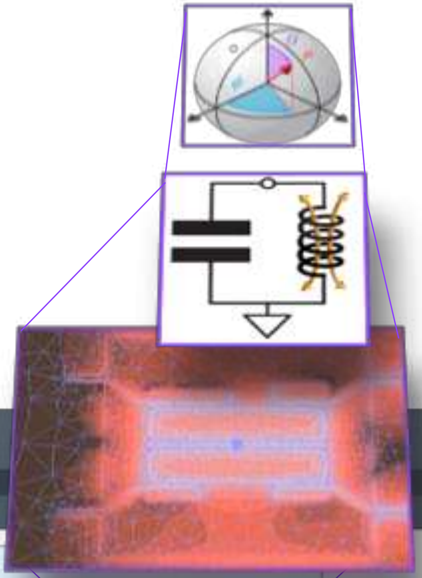
Qiskit | quantum device design
Metal

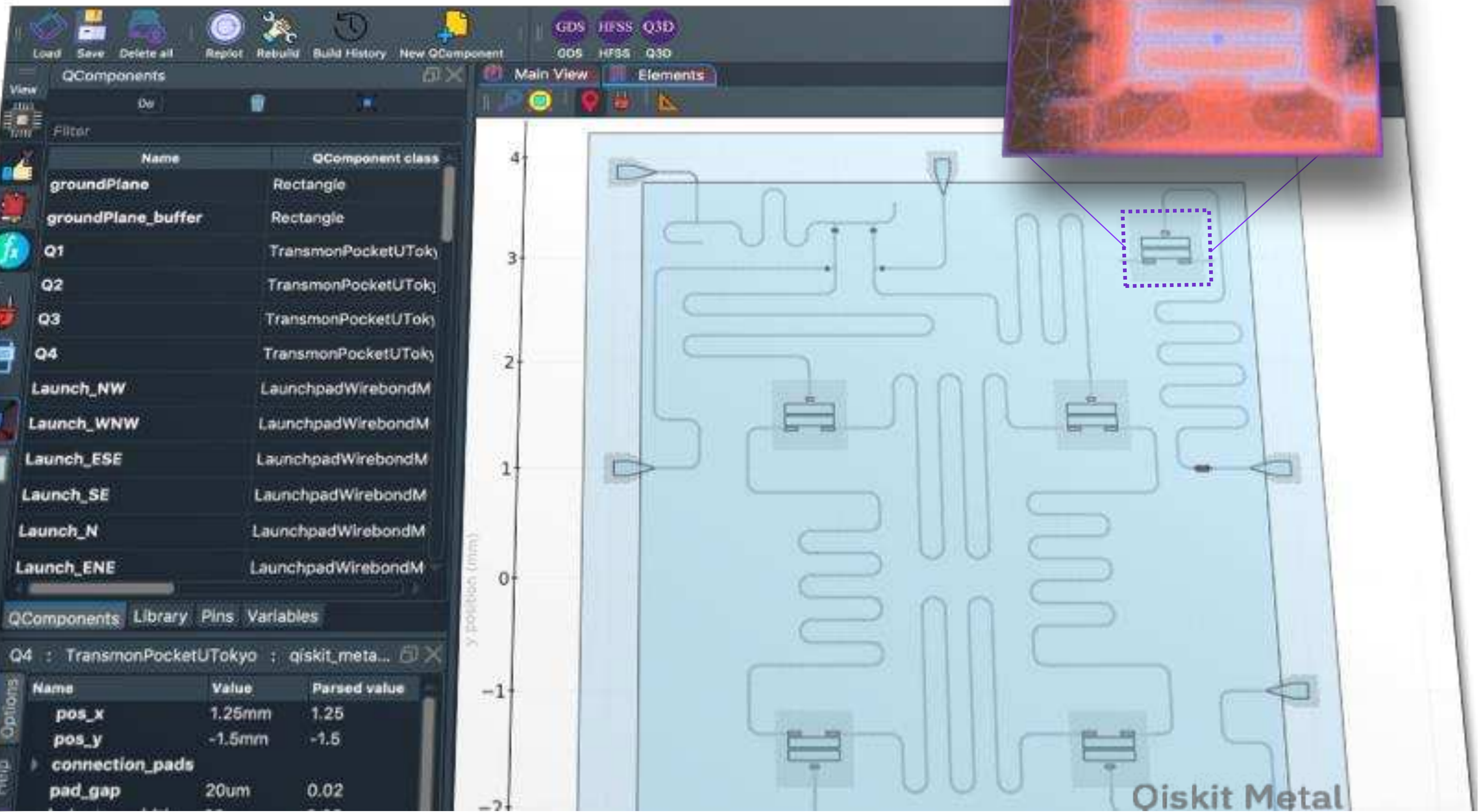
+ advanced talk (today)
+ workshops (tomorrow)
+ lunch meeting discussion (today)

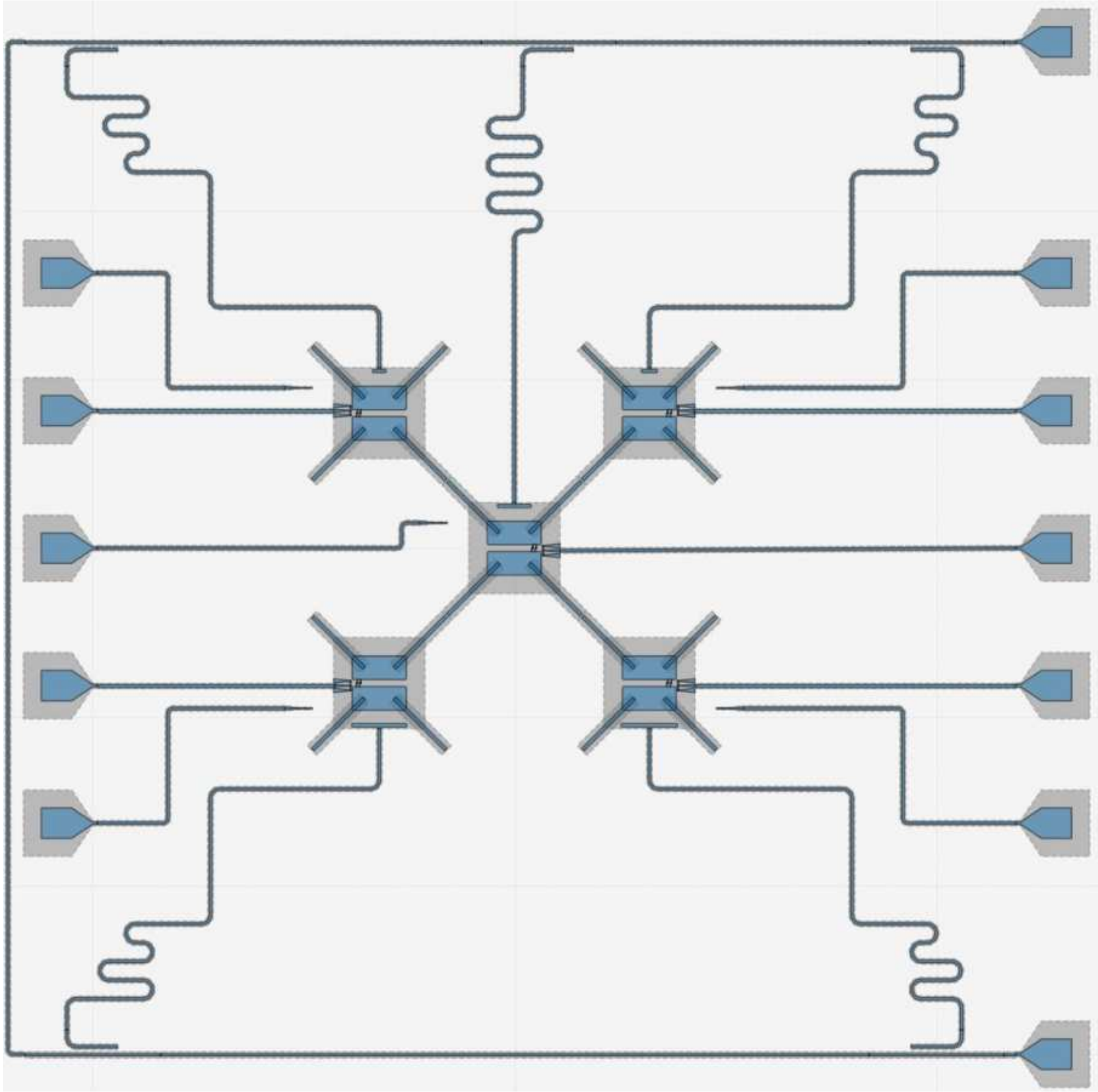
Community ecosystem driven (Apache-2.0)
github.com/qiskit-community/qiskit-metal











Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE

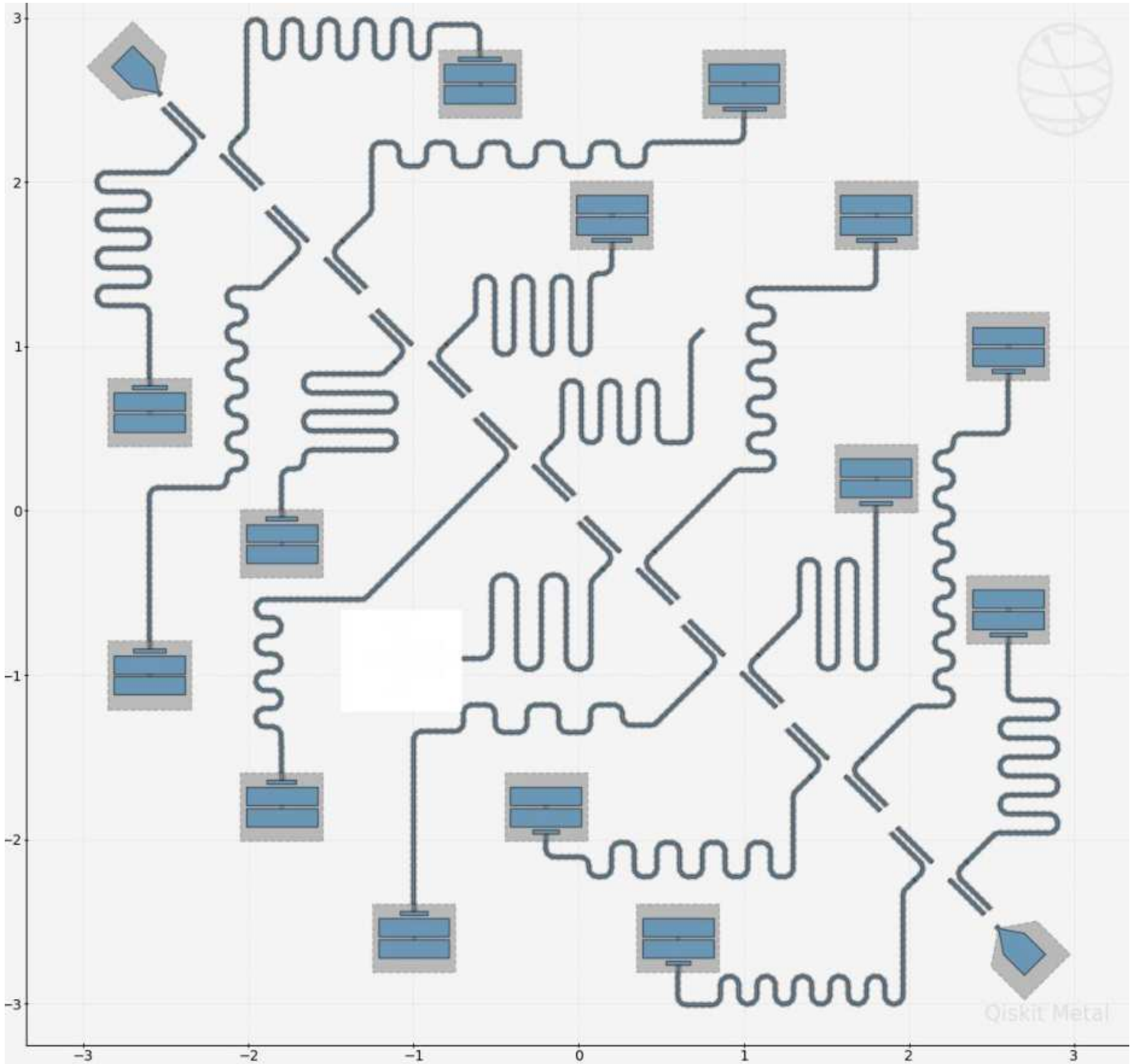
SQuID Lab

Qiskit | quantum device design
Metal

5 tunable transmon weight-4 devices
(direct coupling and Delft/ETH style)

Designs:
- Benjamin Joecker, PhD
- Thue Christian Thann, PhD Candidate

Affiliations:
- Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE

SQuID Lab

Standard candle devices for tracing reproducibility

Designs:
-Thue Christian Thann, PhD Candidate

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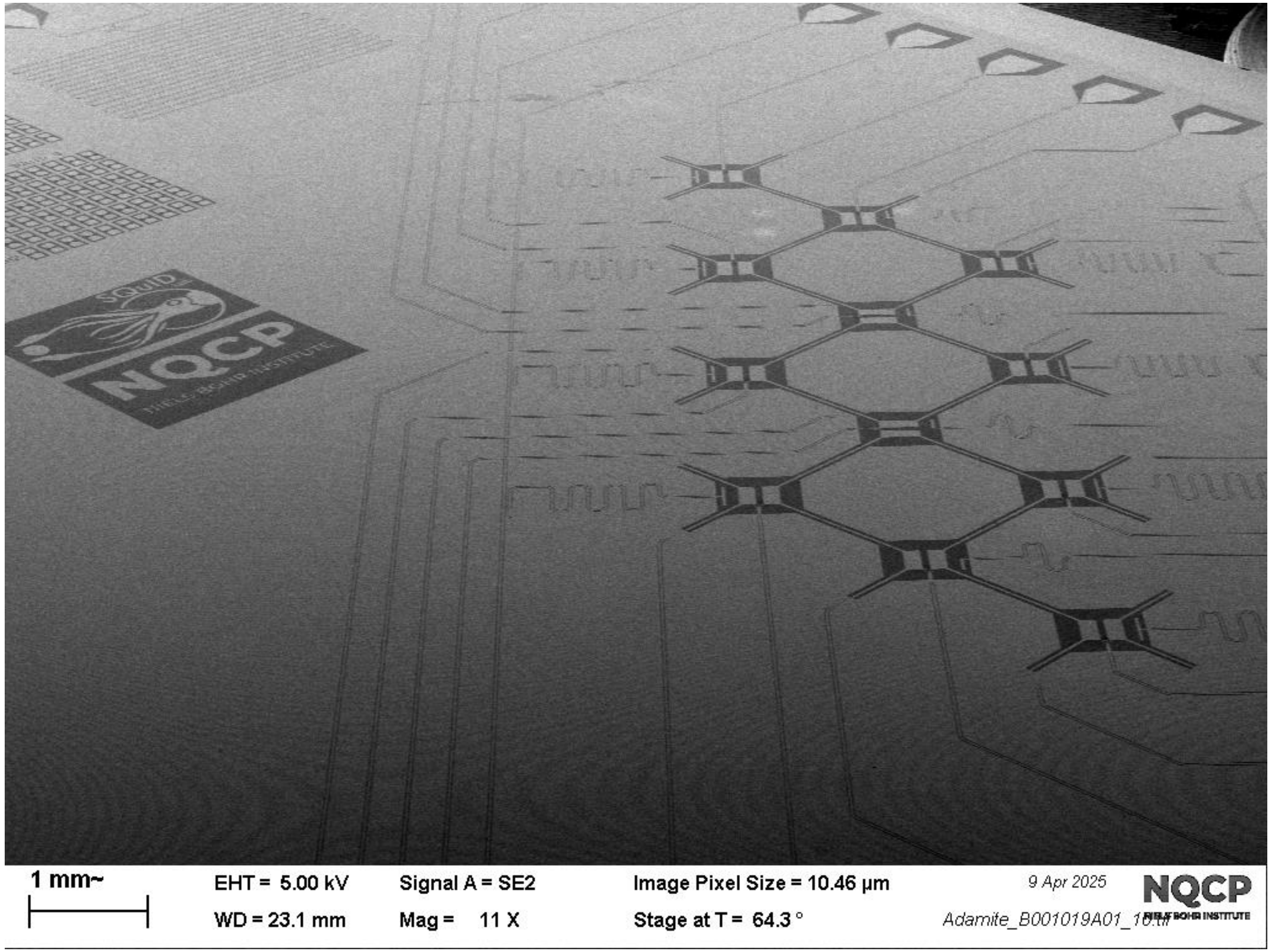
Qiskit | quantum device design
Metal



Example designs from the community, folks like you!

NQCP
NIELS BOHR INSTITUTE

SQuID Lab

Christopher. W. Warren
Niels Bohr Institute
Verified email at nbi.ku.dk
quantum computing quantum simulati
quantum information

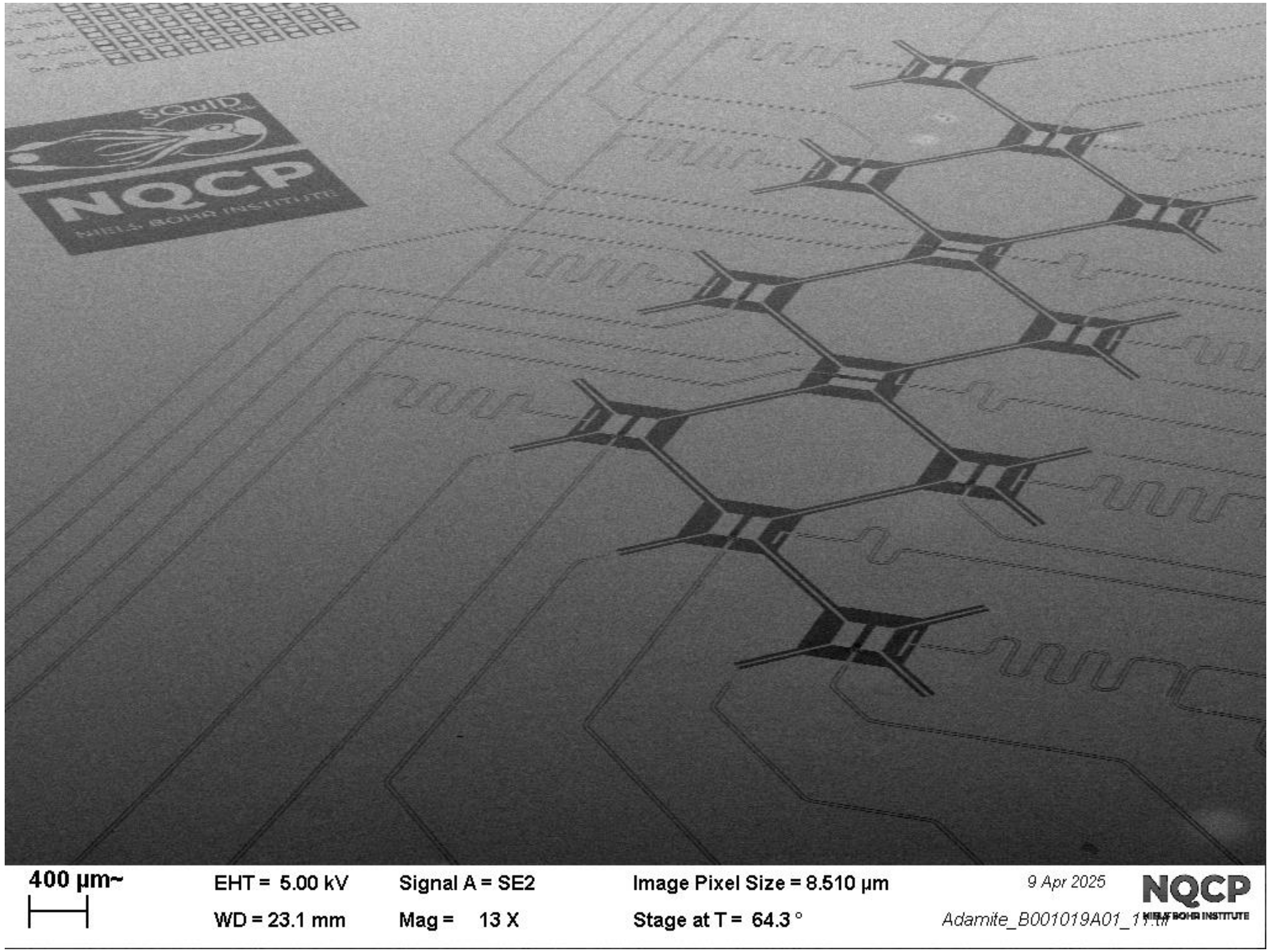
12 Tunable qubit chip; directly coupled (dubbed "Adamite")

Design:
- Christopher W. Warren, PhD

Affiliations:
- SQuID Lab, Niels Bohr Institute, University of Copenhagen
- Novo Nordisk Quantum Computing Programme (NQCP)

Qiskit | quantum device design
Metal

Zlatko Minev



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SQuID Lab

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Qiskit | quantum device design
Metal

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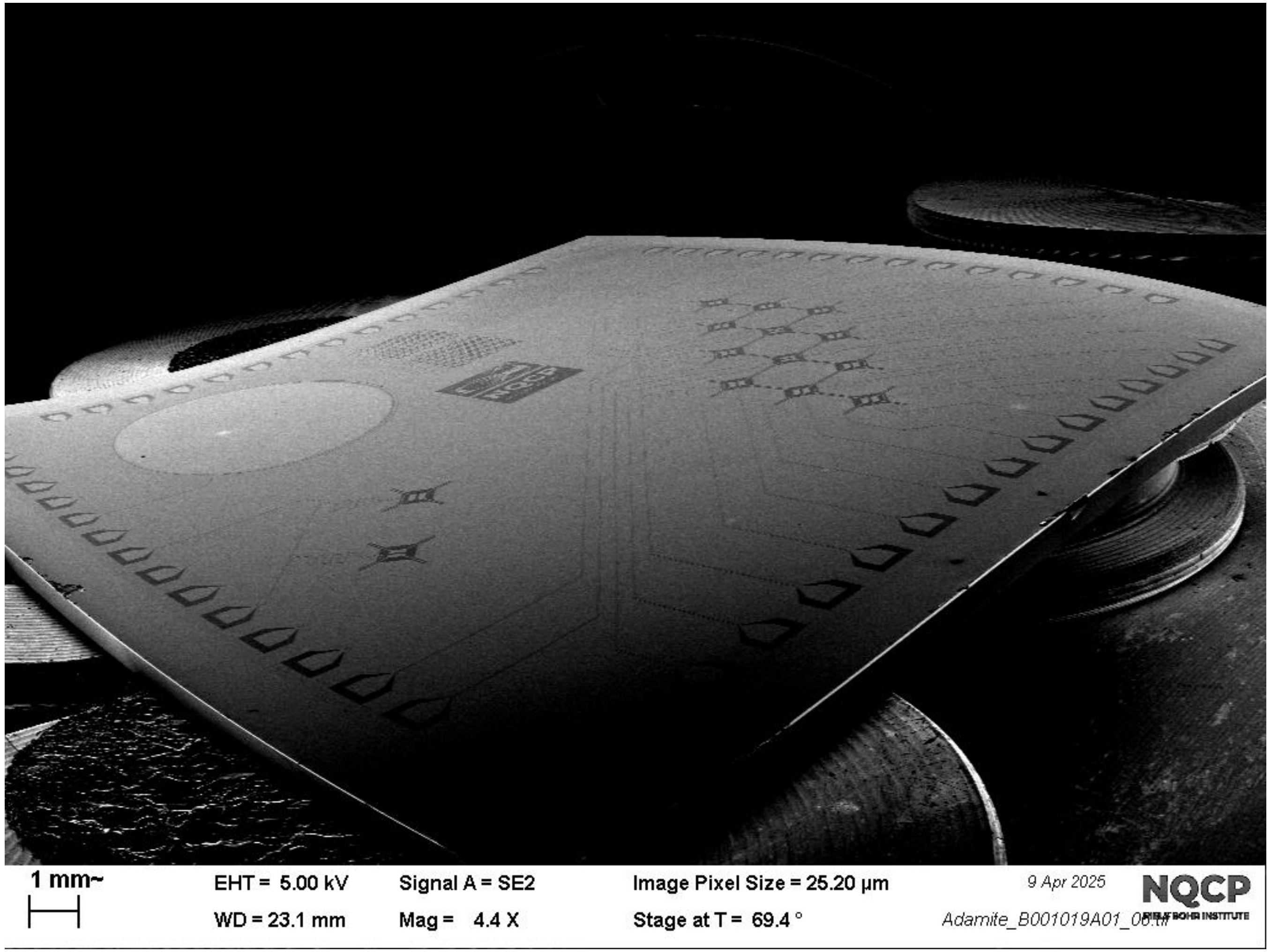
quantum information



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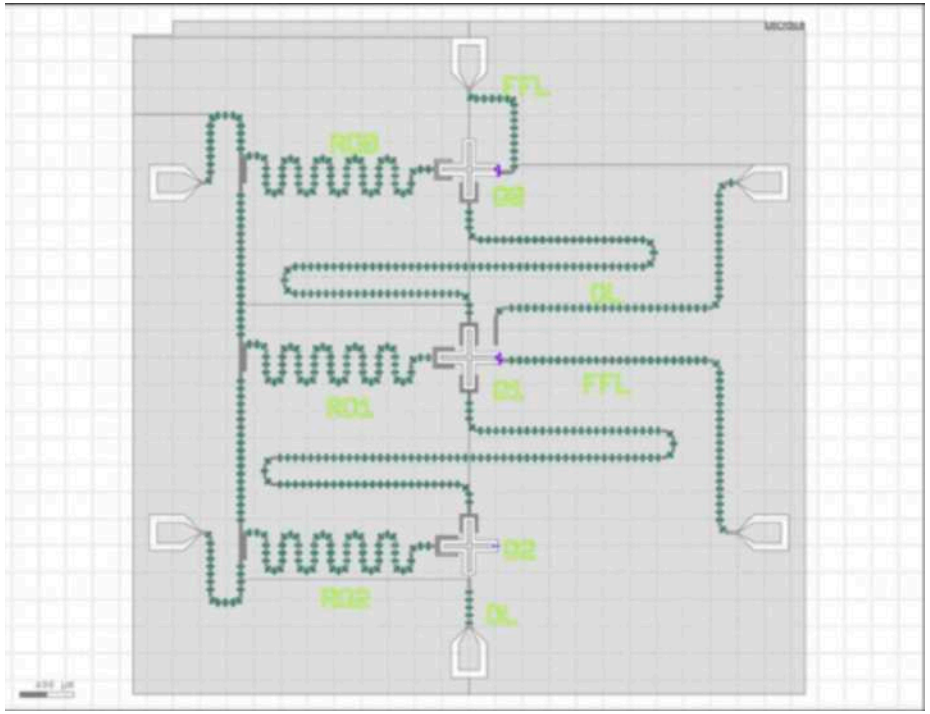
Qiskit | quantum device design
Metal

Zlatko Minev

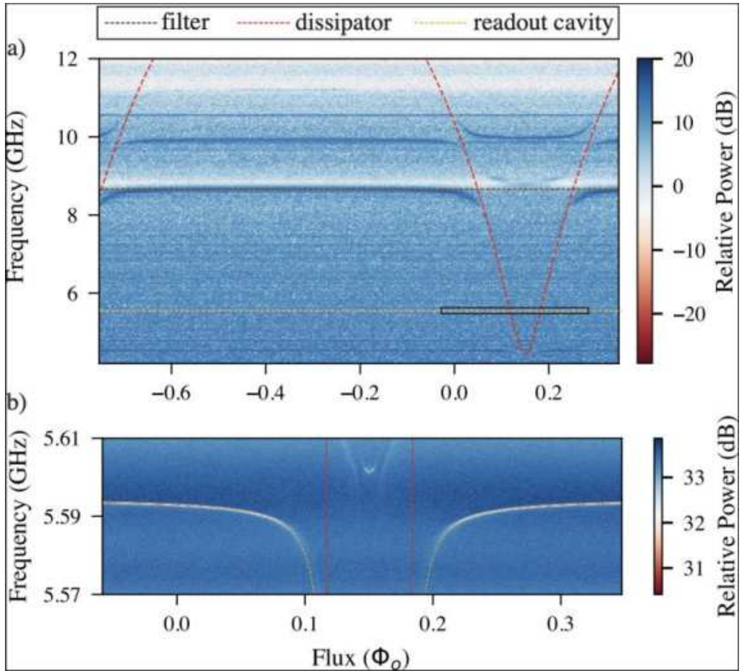


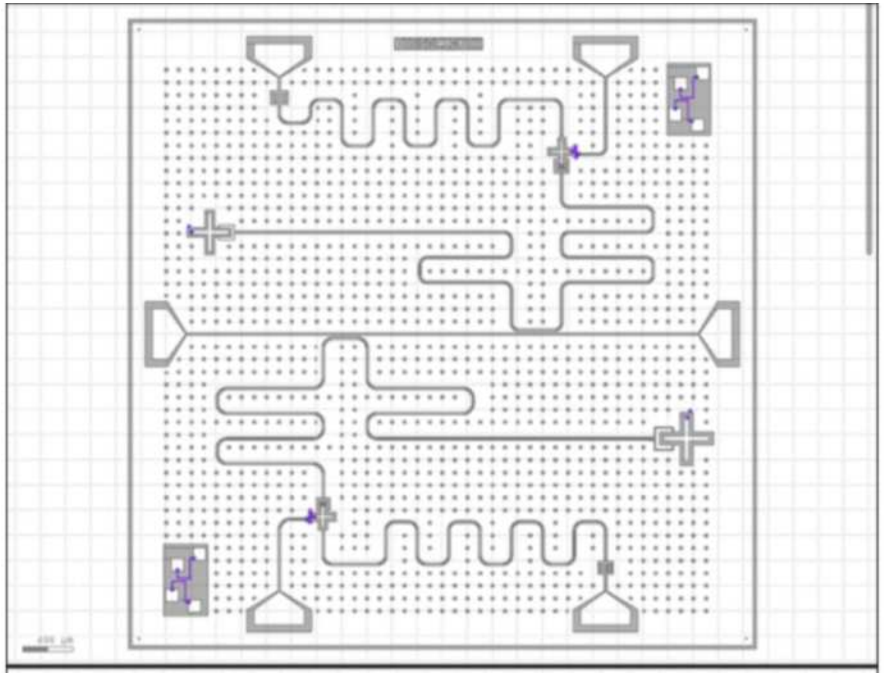
Example designs from the community, folks like you!

USC Viterbi
School of Engineering
Levenson-Falk Lab (LFL)

Sadman Ahmed Shanto

Eli Levenson-Falk





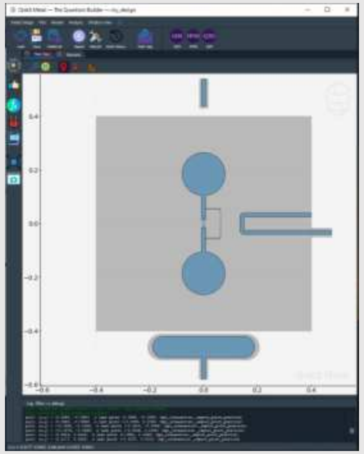
Qiskit | quantum device design
Metal

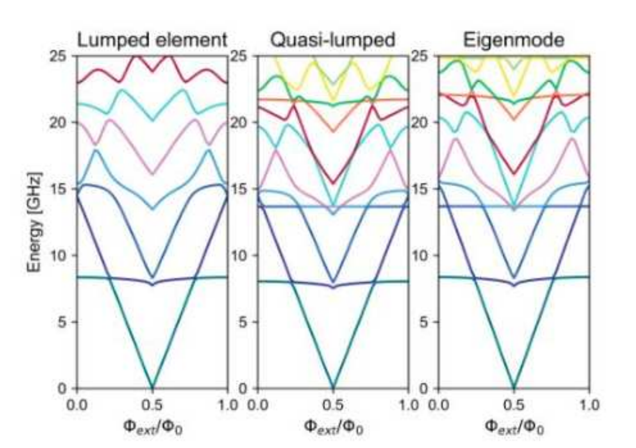
Zlatko Minev

Example designs from the community, folks like you!

QuTech / **Technische Universiteit Delft**

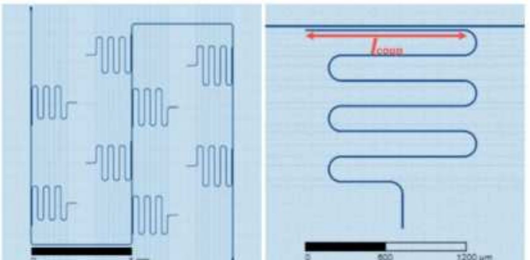
Figen Yilmaz, C. Andersen Lab: Fluxonium chip design
Roald van den Boogaart-Andersen Lab: Fluxonium spectrum (eigenmode = pyEPR) to show how they extended the analysis methods to be able to handle fluxonium qubits
Sara Buhktari-Andersen Lab: Fabricated resonator design being measured





3.1. Final design

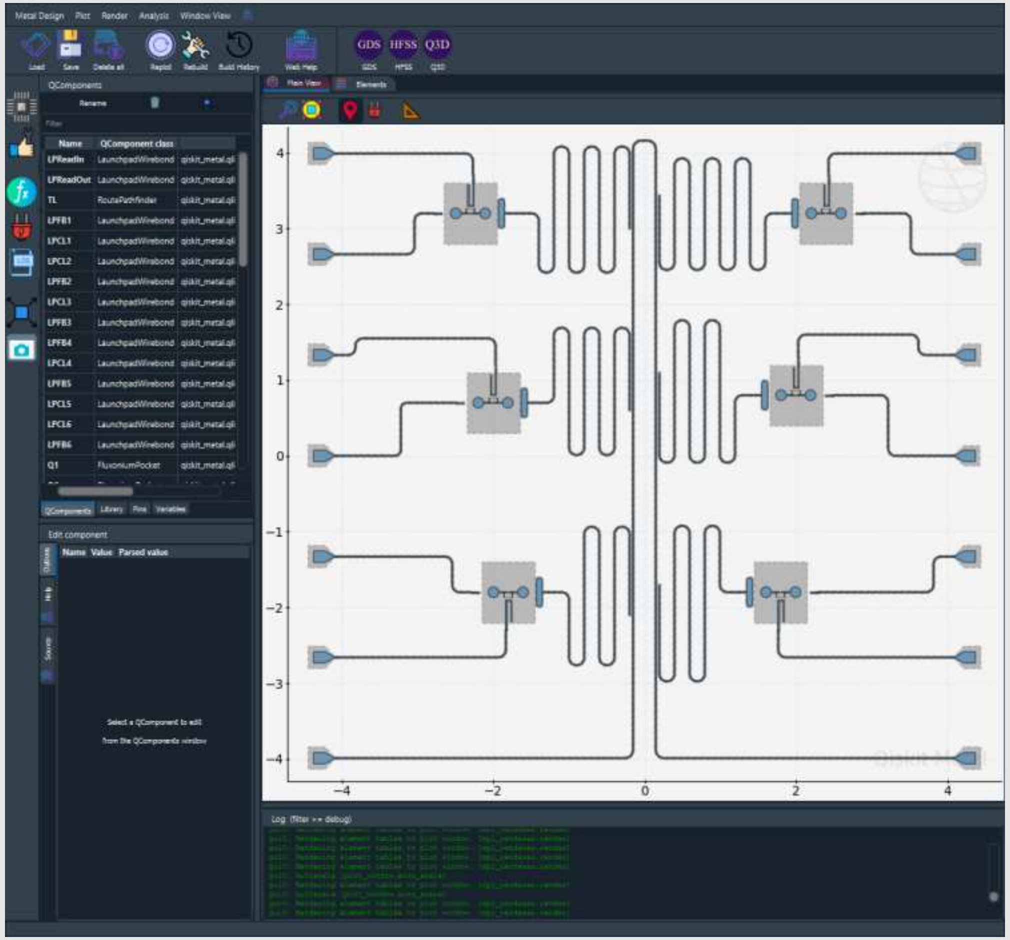
The final design consists of an S-shaped feedline routed between two launchpads. Eight resonators are capacitively coupled to this feedline. Every component on this 9 x 9 mm device is implemented by a coplanar waveguide transmission line. The schematic representations of the full device and single resonator geometry are depicted in Figure 6.



"Qiskit-Metal is the best tool if one studies superconducting qubits" -- Figen Yilmaz

QuTech / **Technische Universiteit Delft**

Intermediate chip design.



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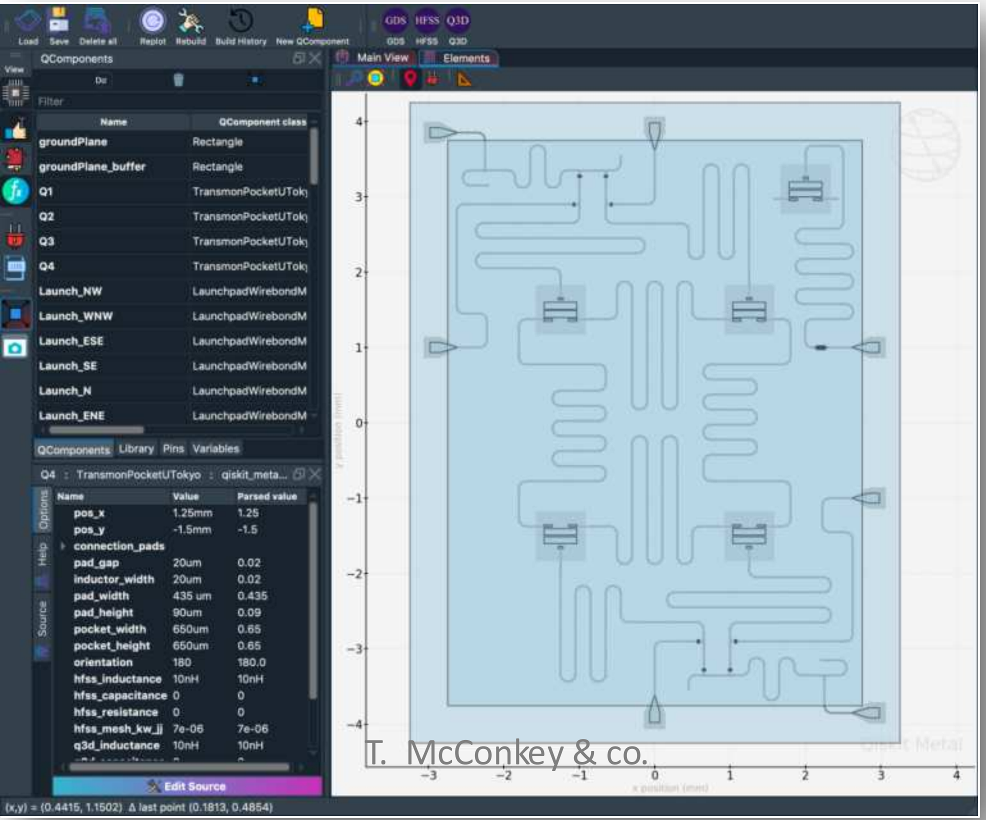
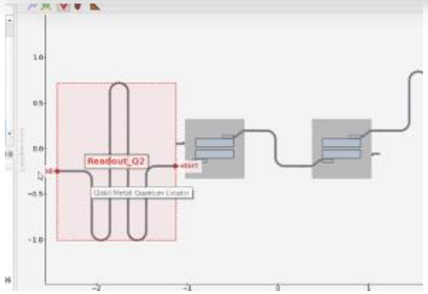
Example designs from the community, folks like you!

IBM 5Q Tsuru U Tokyo



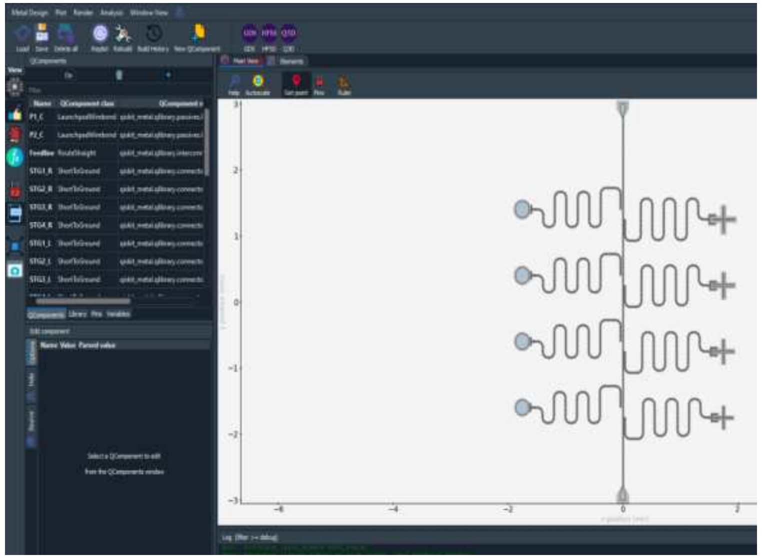
T. McConkey & co.



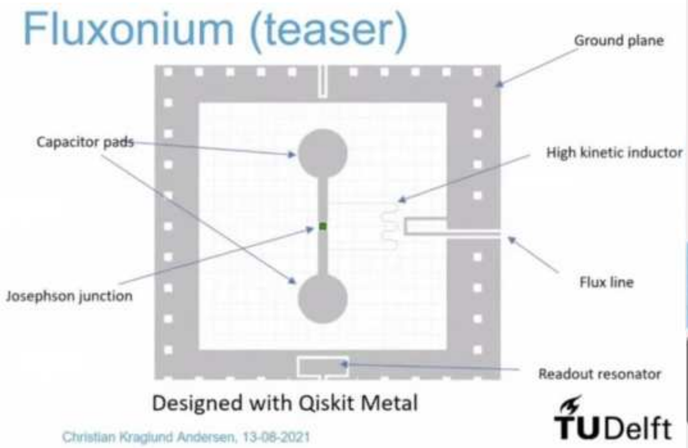
APS physics MARCH MEETING 2021 MARCH 15-19 ONLINE

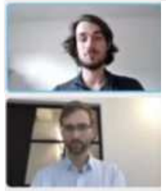
CEC/ICMC 21 VIRTUAL CONFERENCE JULY 19-23 Cryogenics·Through·Collaboration

Chalmers 8Q

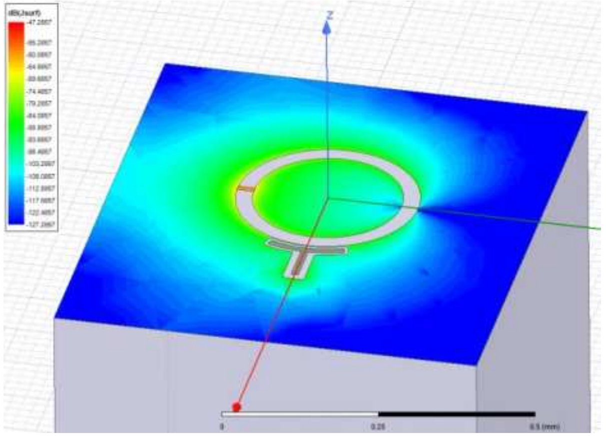


TU Delft Fluxonium

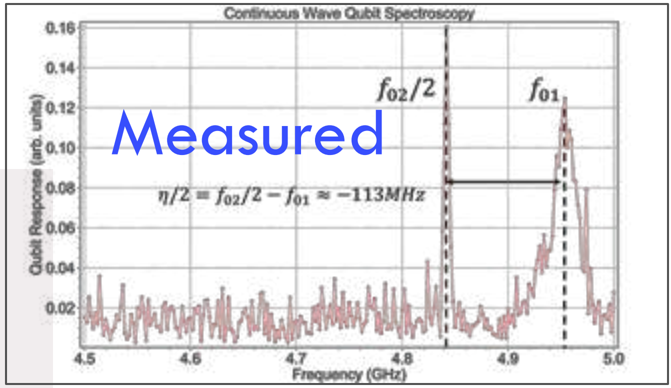




Simulated



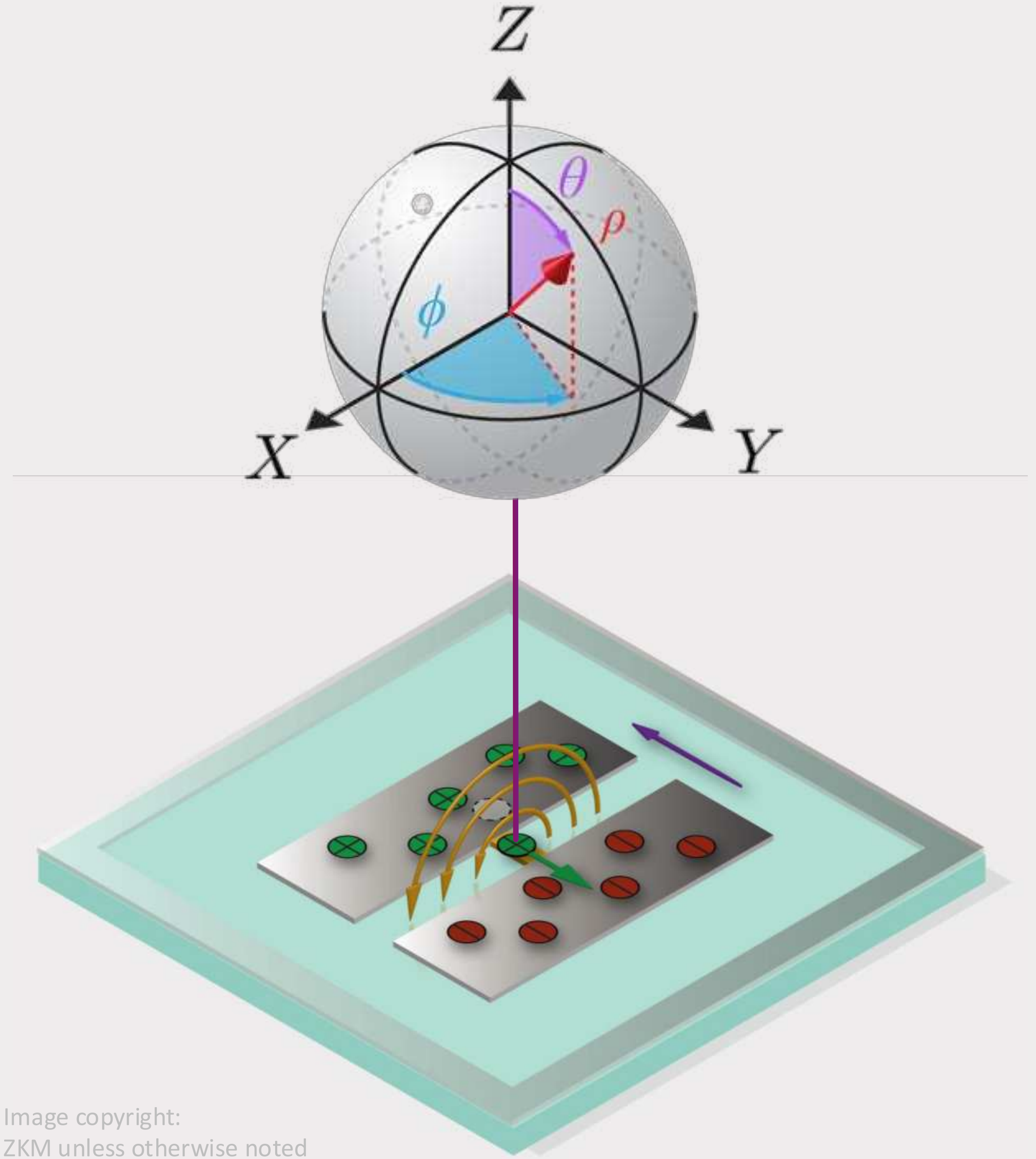
Measured



...

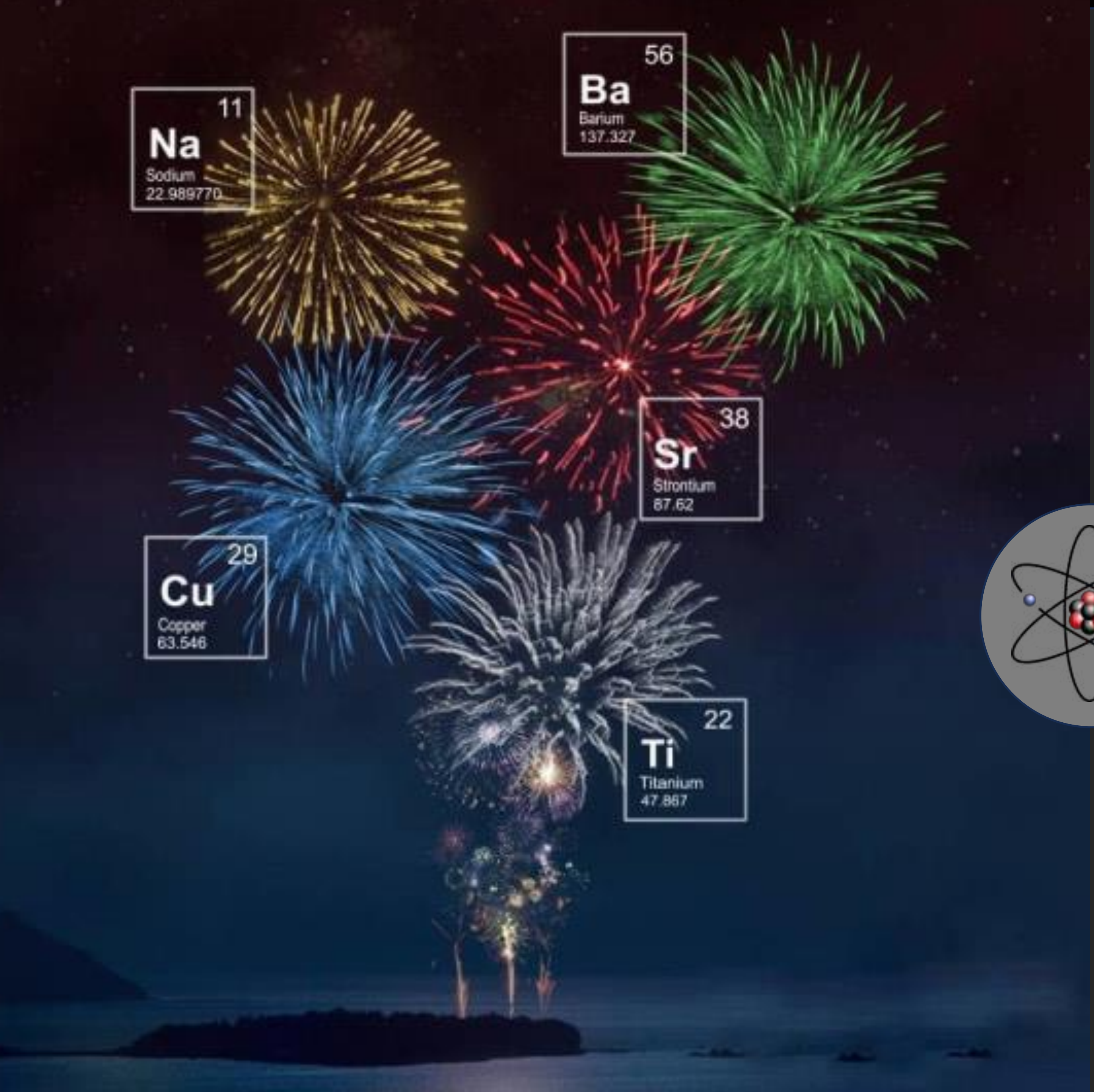
IEEE QUANTUM WEEK IEEE International Conference on Quantum Computing and Engineering — QCE21

Zlatko Minev



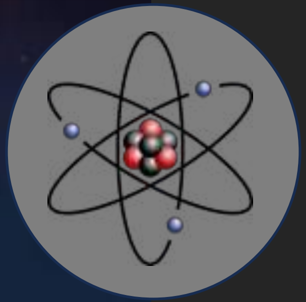
Quantum bit

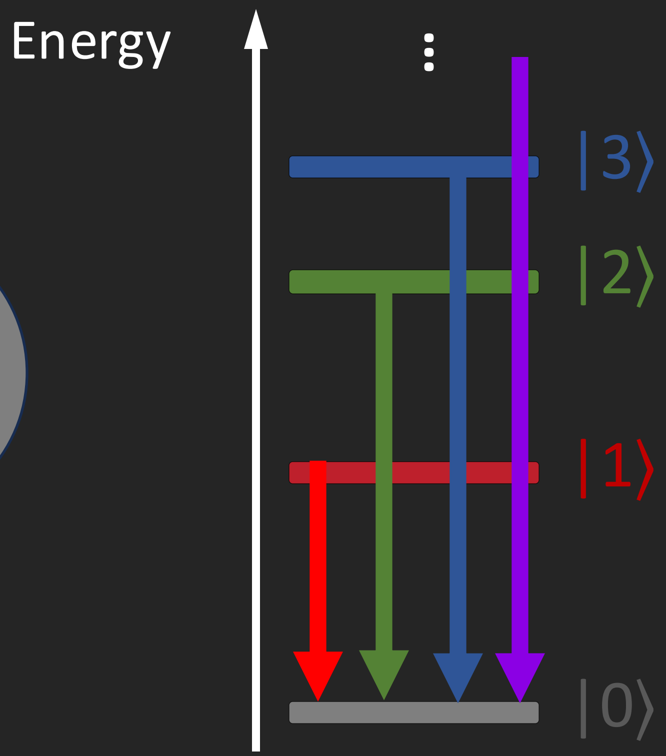
Reality to Abstraction



The Light of Atoms

Atoms are quantum:
discrete intrinsic energy levels*





Quantization of energy

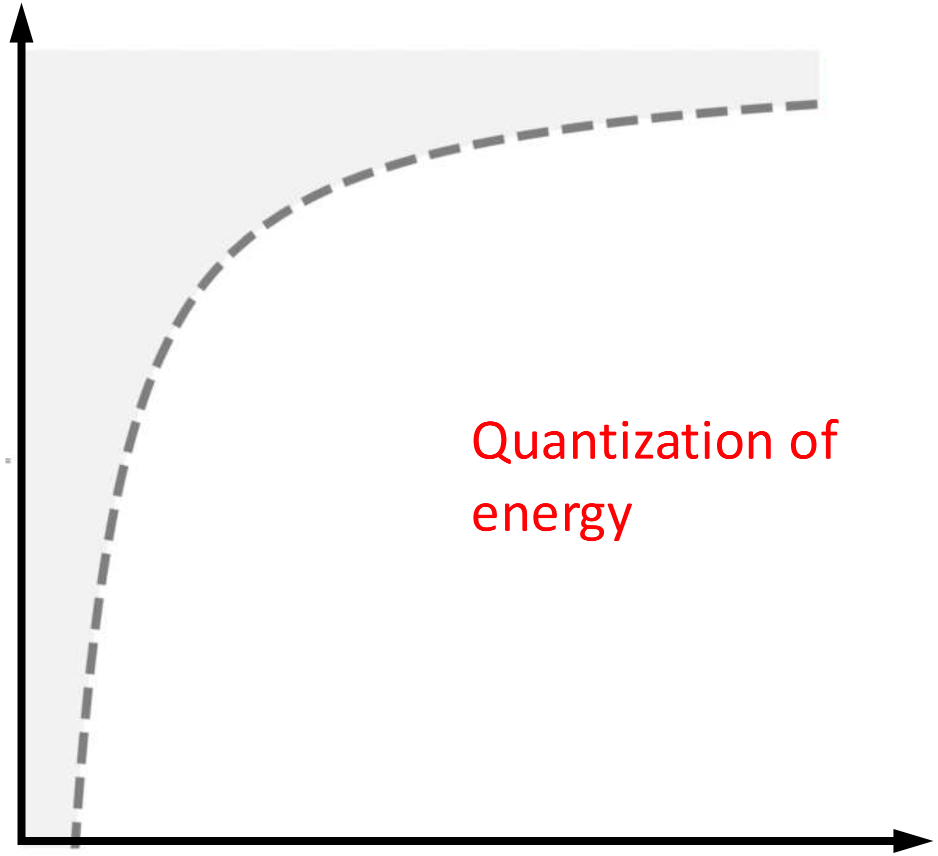
The notion of an energy level was proposed by Bohr in 1913.

Image source msnbc

Zlatko Minev

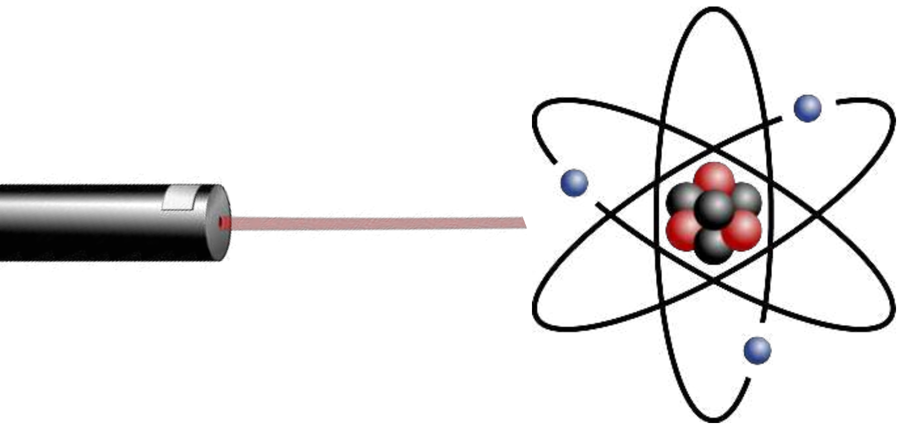
Atomic energy levels and transitions

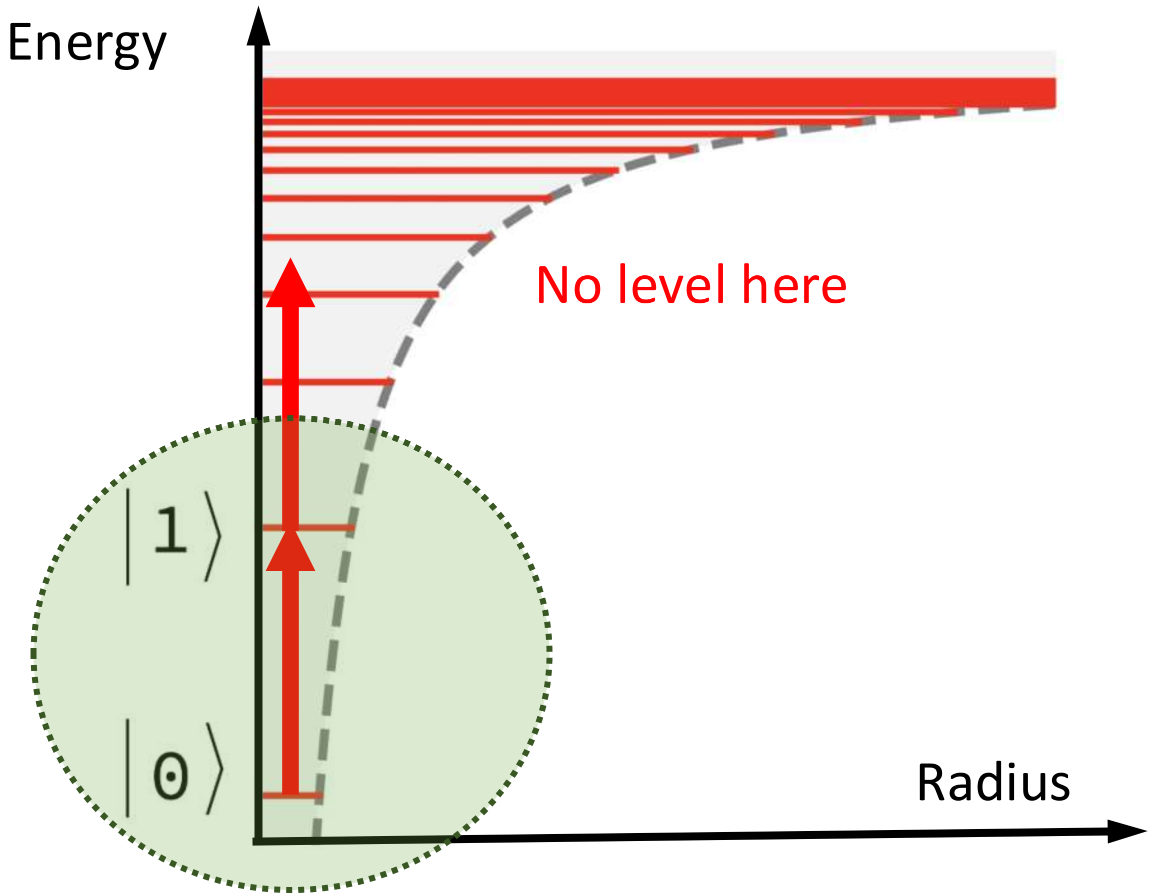
Electron potential-energy landscape



Radial distance from nucleus

Qubit from Atom





Anharmonic degree of freedom and spin

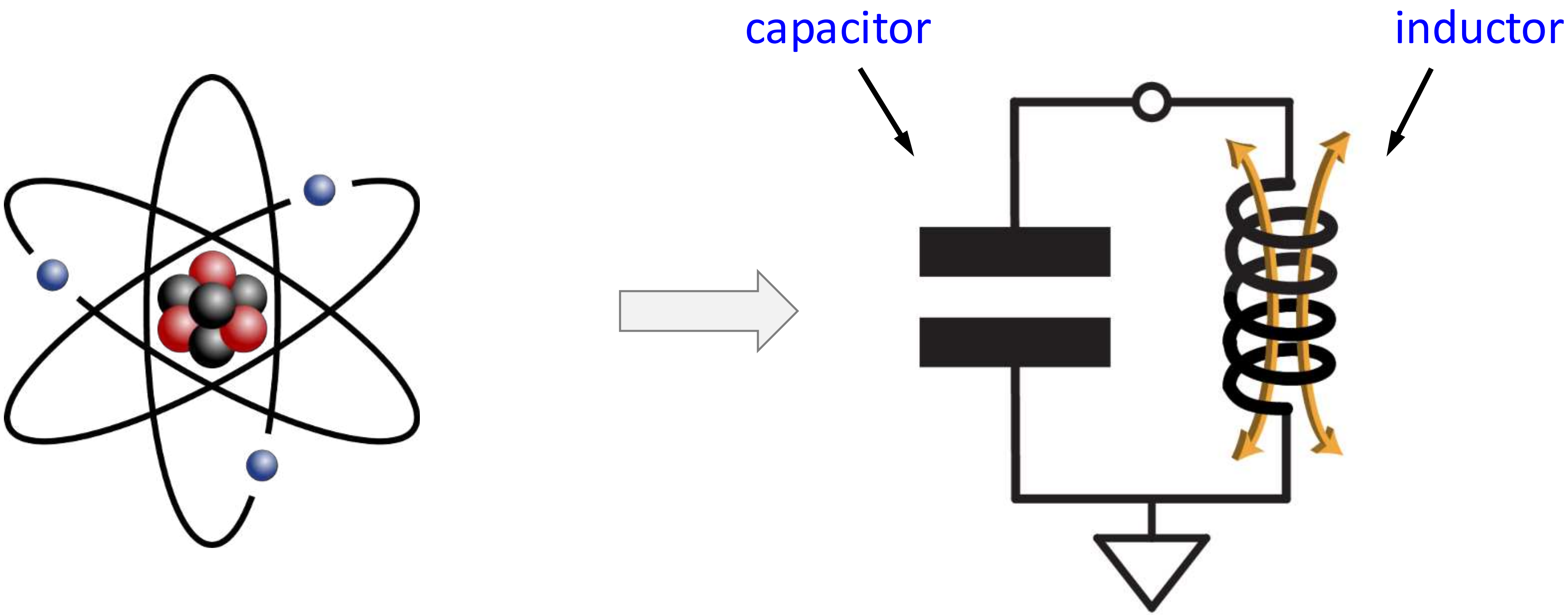
Isolated from environment and thermal bath

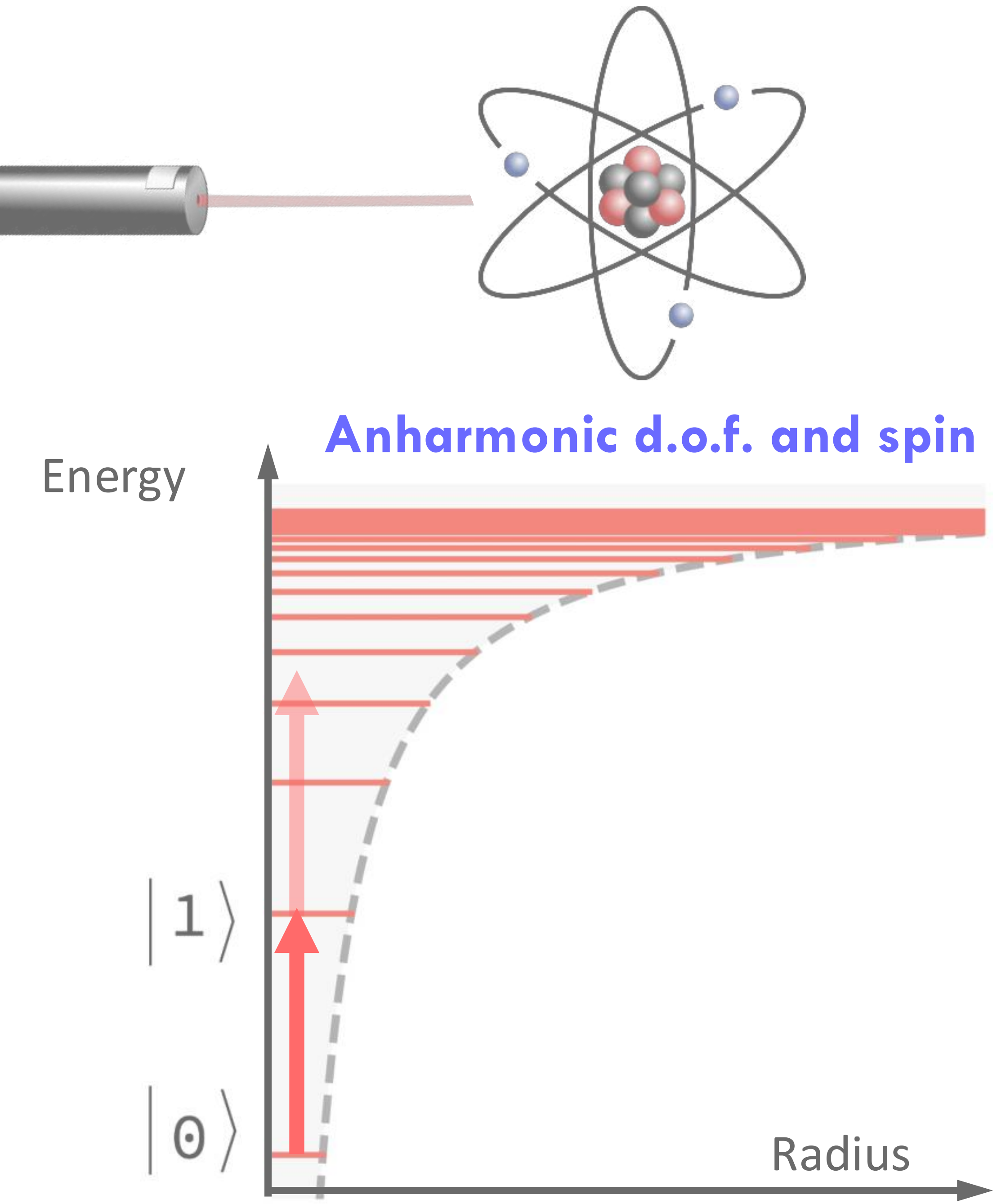
Low-loss

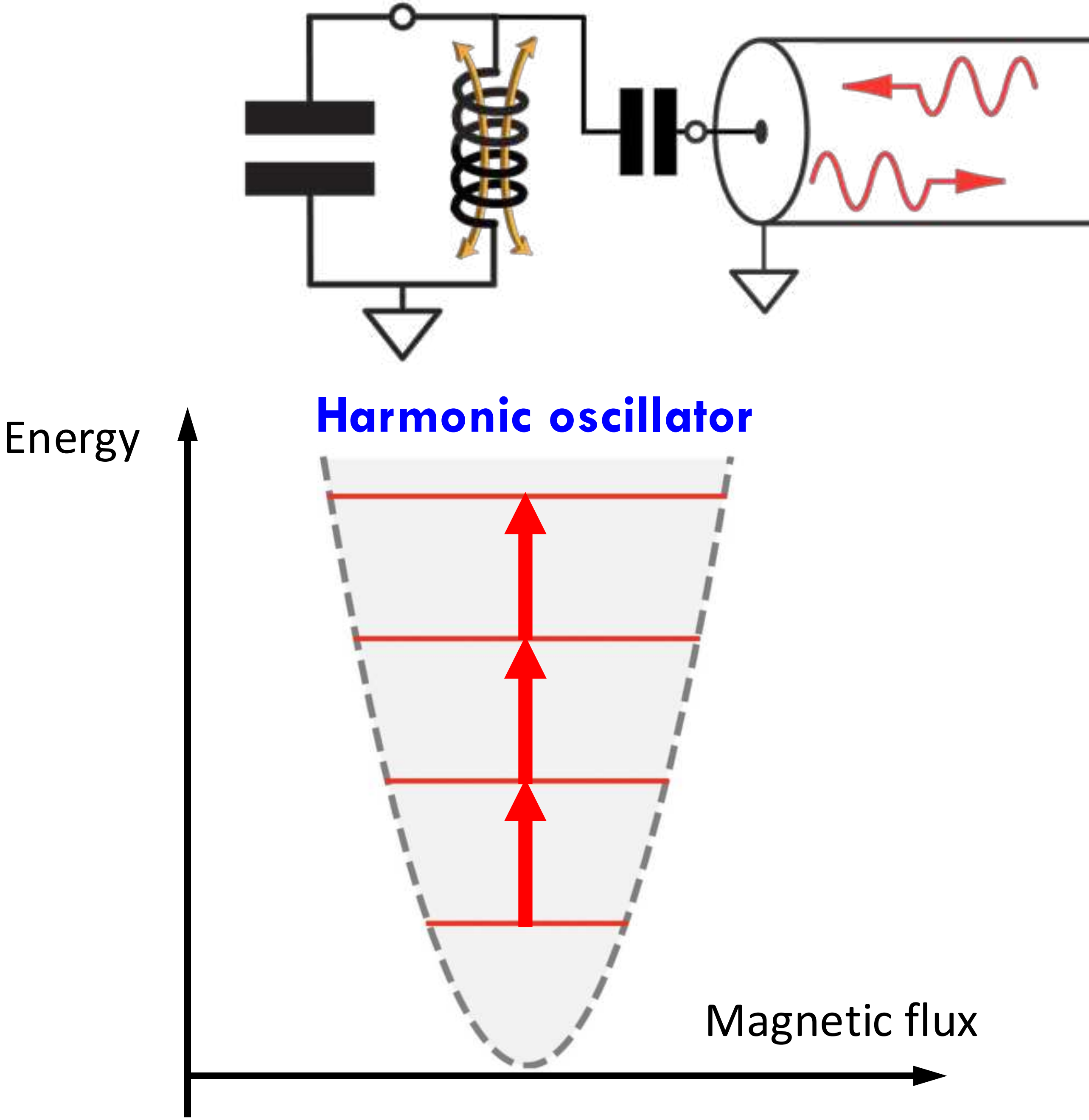
Level diagram allows for qubit-specific control and readout

There are always more than two levels

Artificial atoms







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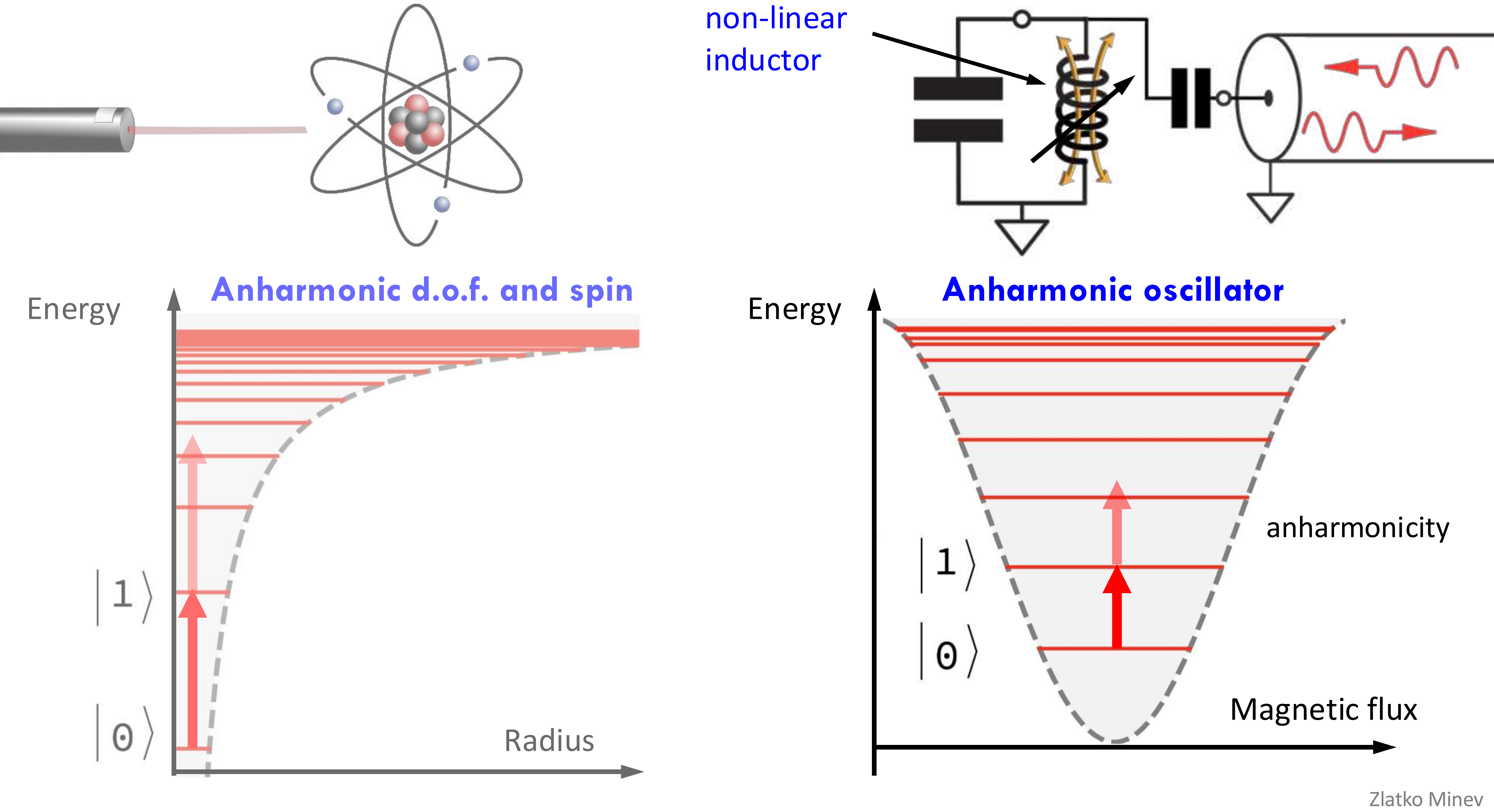

non-linear
inductor
Anharmonic d.o.f. and spin
Energy
|1⟩
|0⟩
Radius
Anharmonic oscillator
Energy
anharmonicity
|1⟩
|0⟩
Magnetic flux
Zlatko Minev

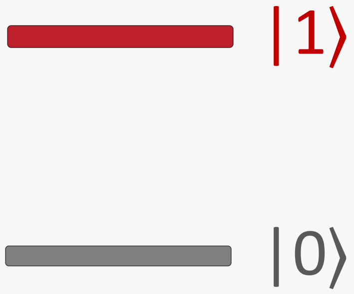
Big-picture connections

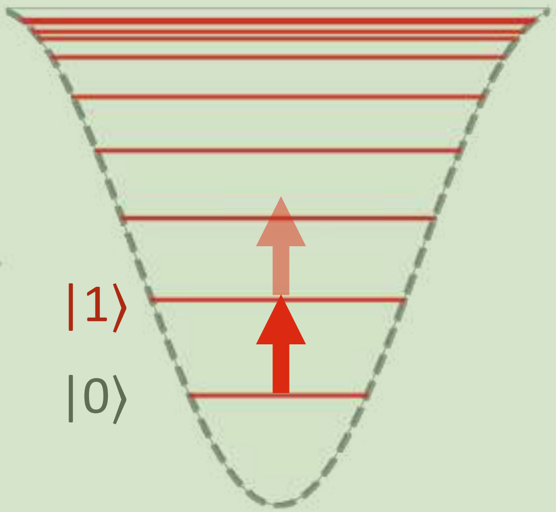
Idealization of *qubit*

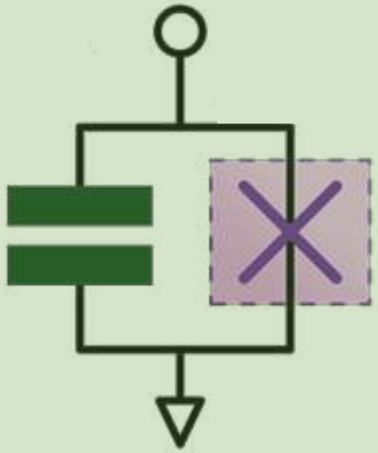
Anharmonic *oscillator*

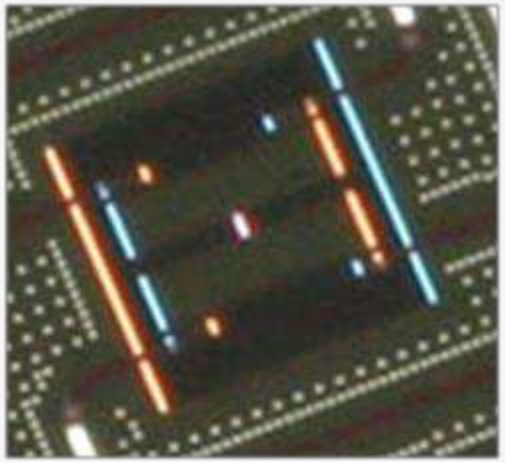
Physical circuit model

Physical layout









Idealization → **Physical reality**

Circuit Quantum Electrodynamics (cQED)

Macro
overview

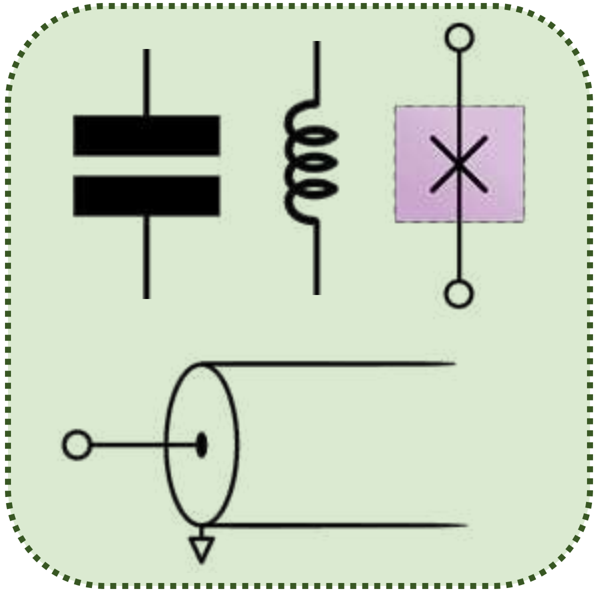
There are two kinds of physicists:

Those who believe all of physics is *spins*.

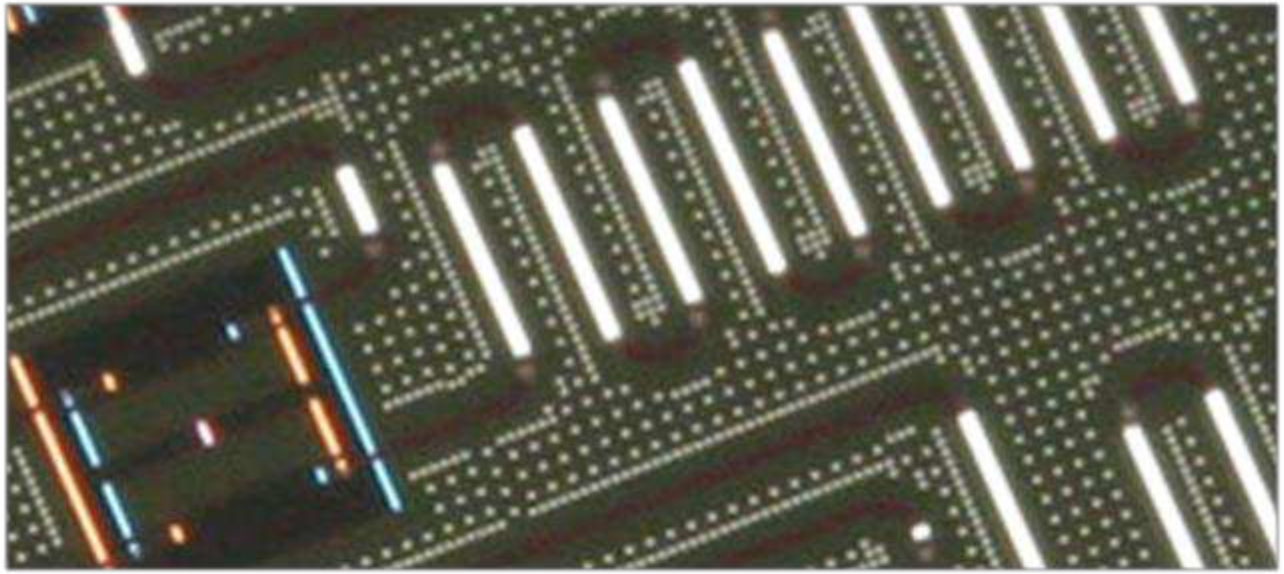
Those who believe all of physics is *oscillators*.

cQED Ingredients

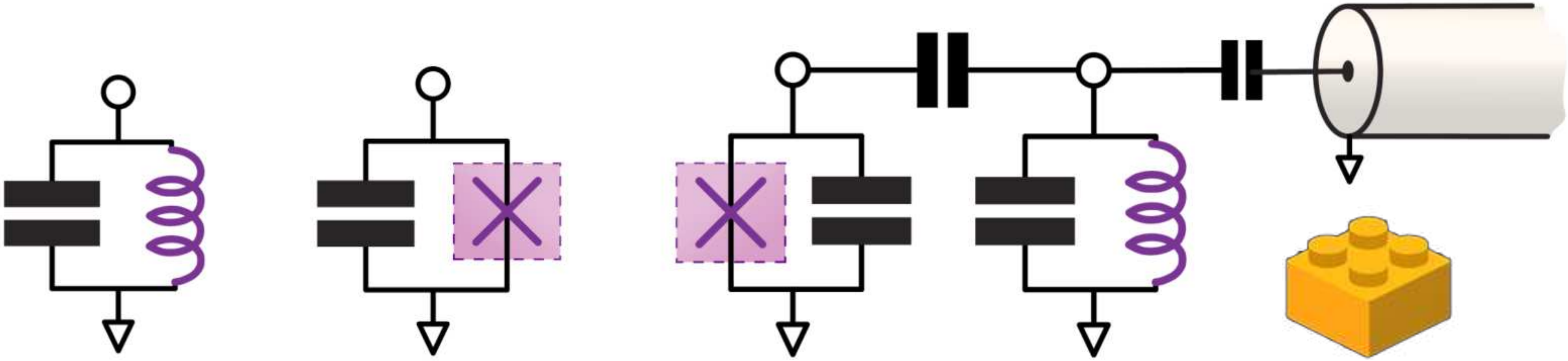
Circuit elements



Microwave oscillators



Combine



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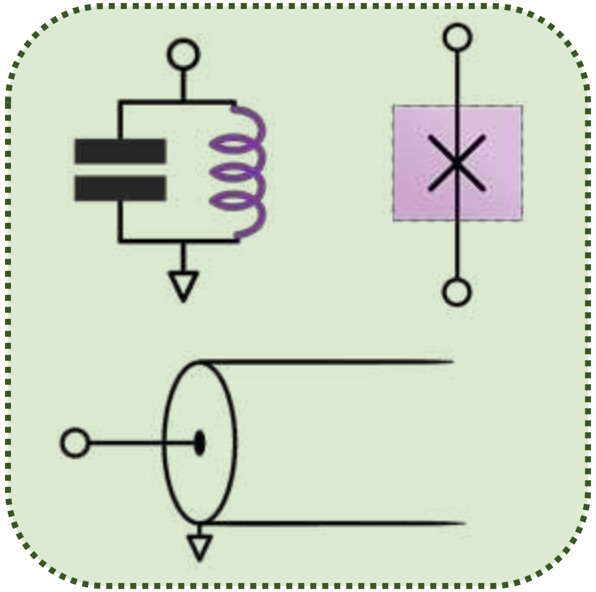
cQED Ingredients



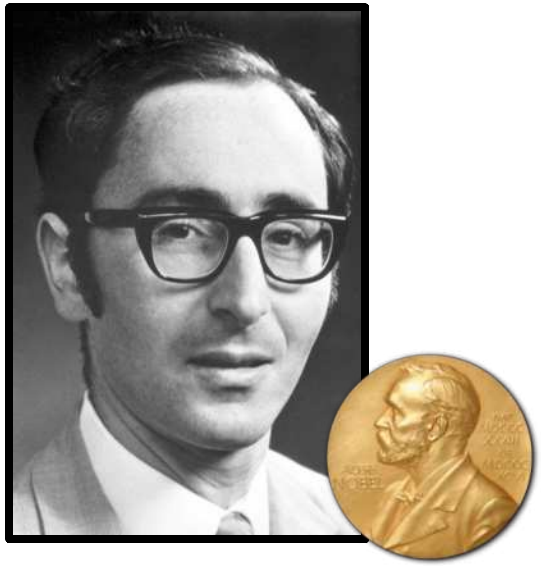
Small dissipation

Isolation from environment

Low temperature

Nonlinearity

Large vacuum fluctuations



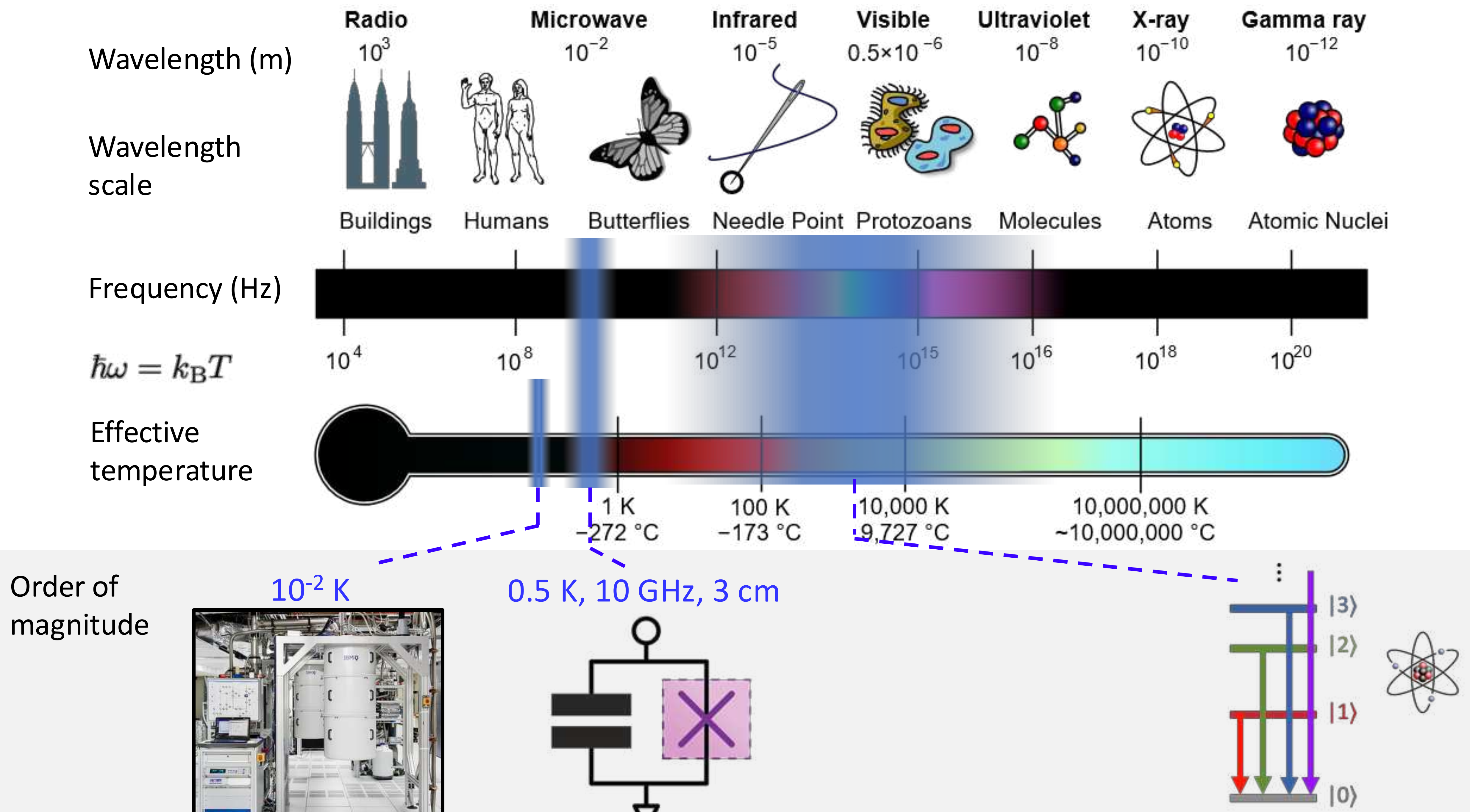
Superconductivity

Nominally zero intrinsic dissipation and heat

Nominal temperature far below energy level splitting

Non-linear, robust Josephson tunnel junction effect

* Low energy dynamics. Microwave image: vectorpocket; Nobel photograph: Jonathunder; Josephson photograph: Nobel Foundation archive.



Spectrum image: Inductiveload, NASA

A few introductory reviews

And many more… check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, … Oliver, W. D. *(2020)*
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., … & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, *6*(2), 021318

Corcoles, A. D., Kandala, A., … Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, *80*(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, *3*(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, *113*, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, *82*(2), 1155–1208

Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, *453*(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

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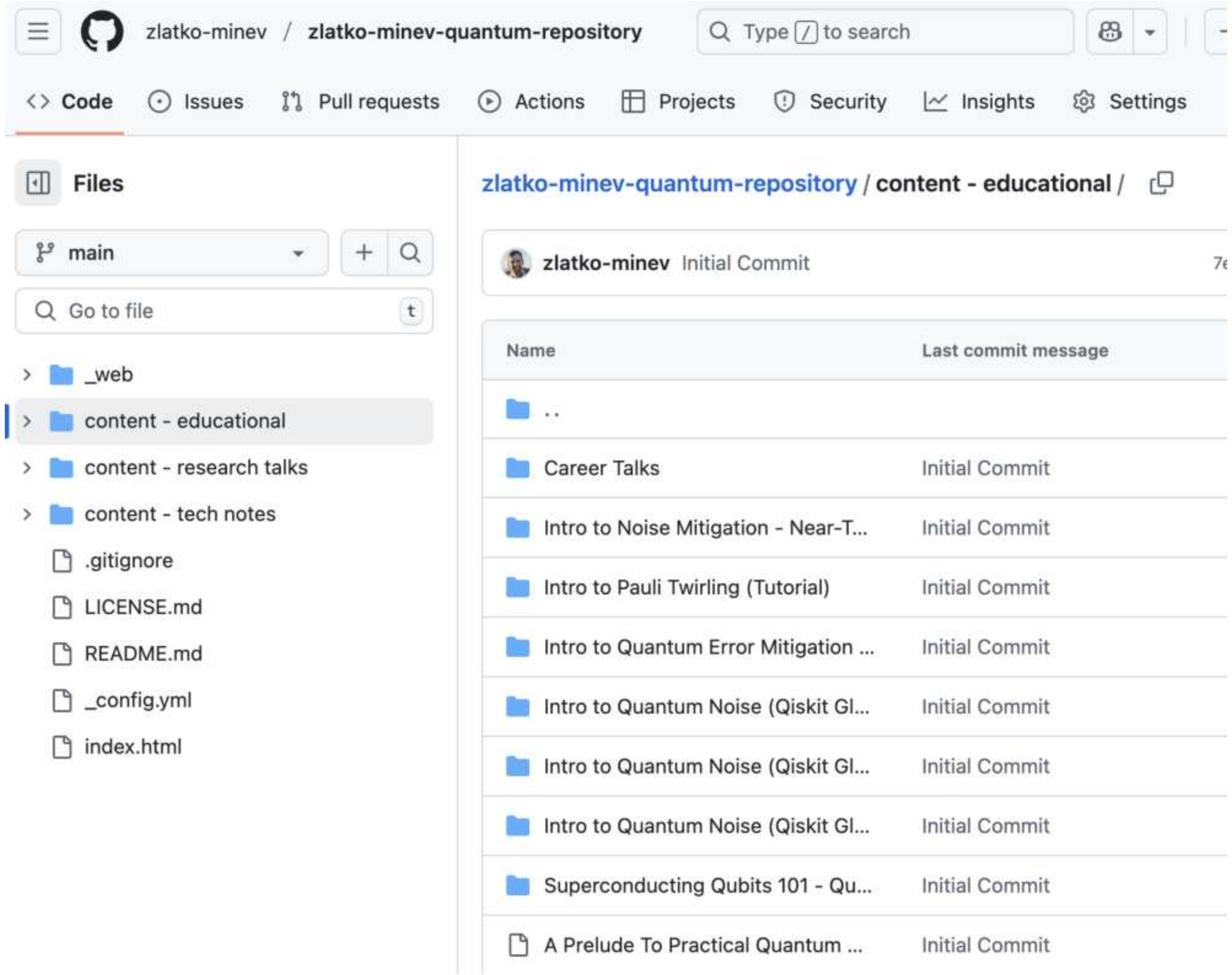
Where are all my slides?

Slides!



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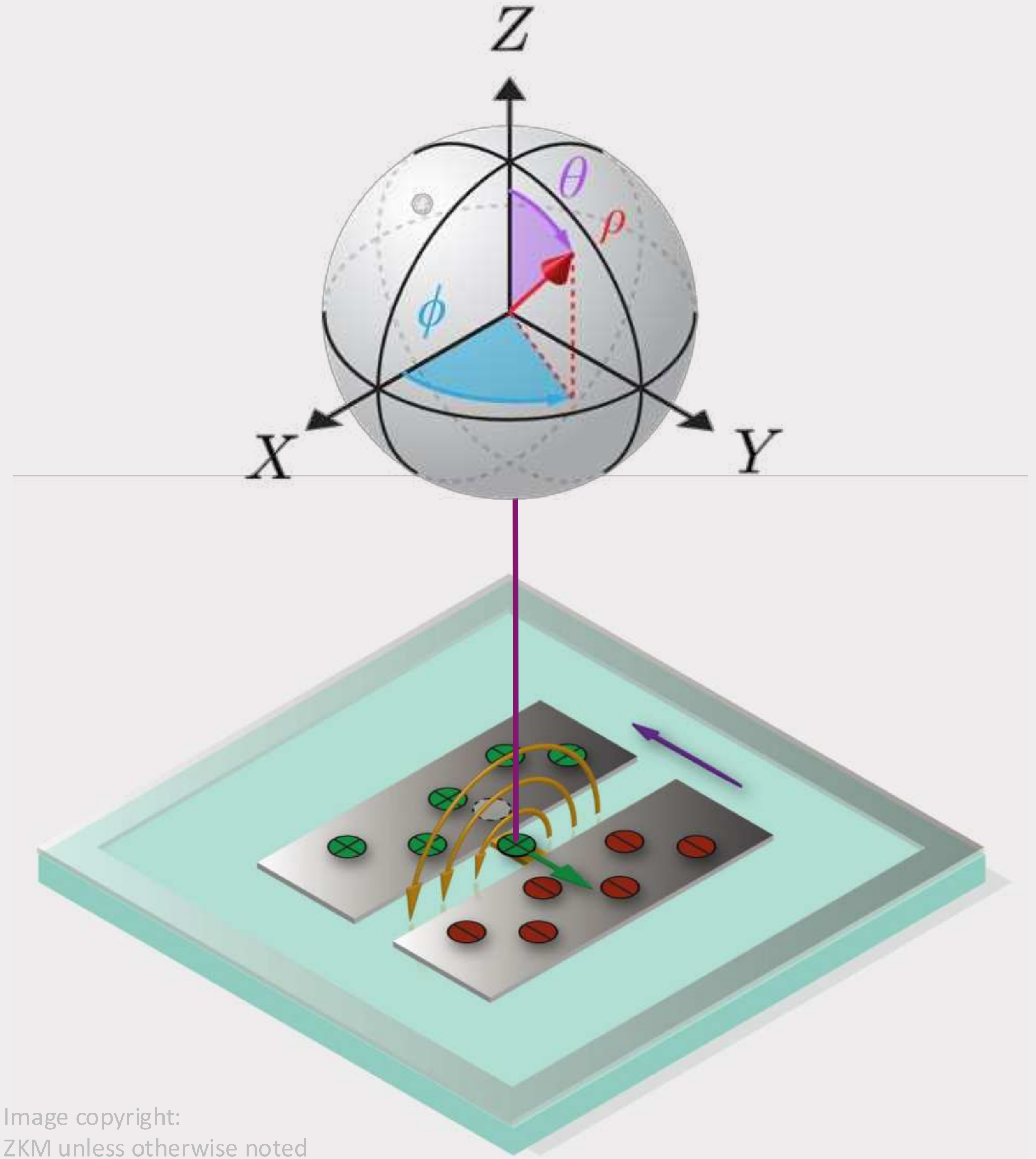
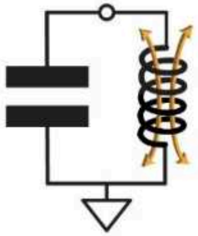
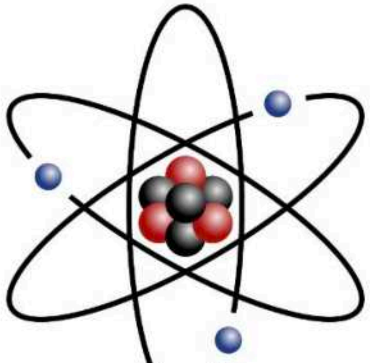
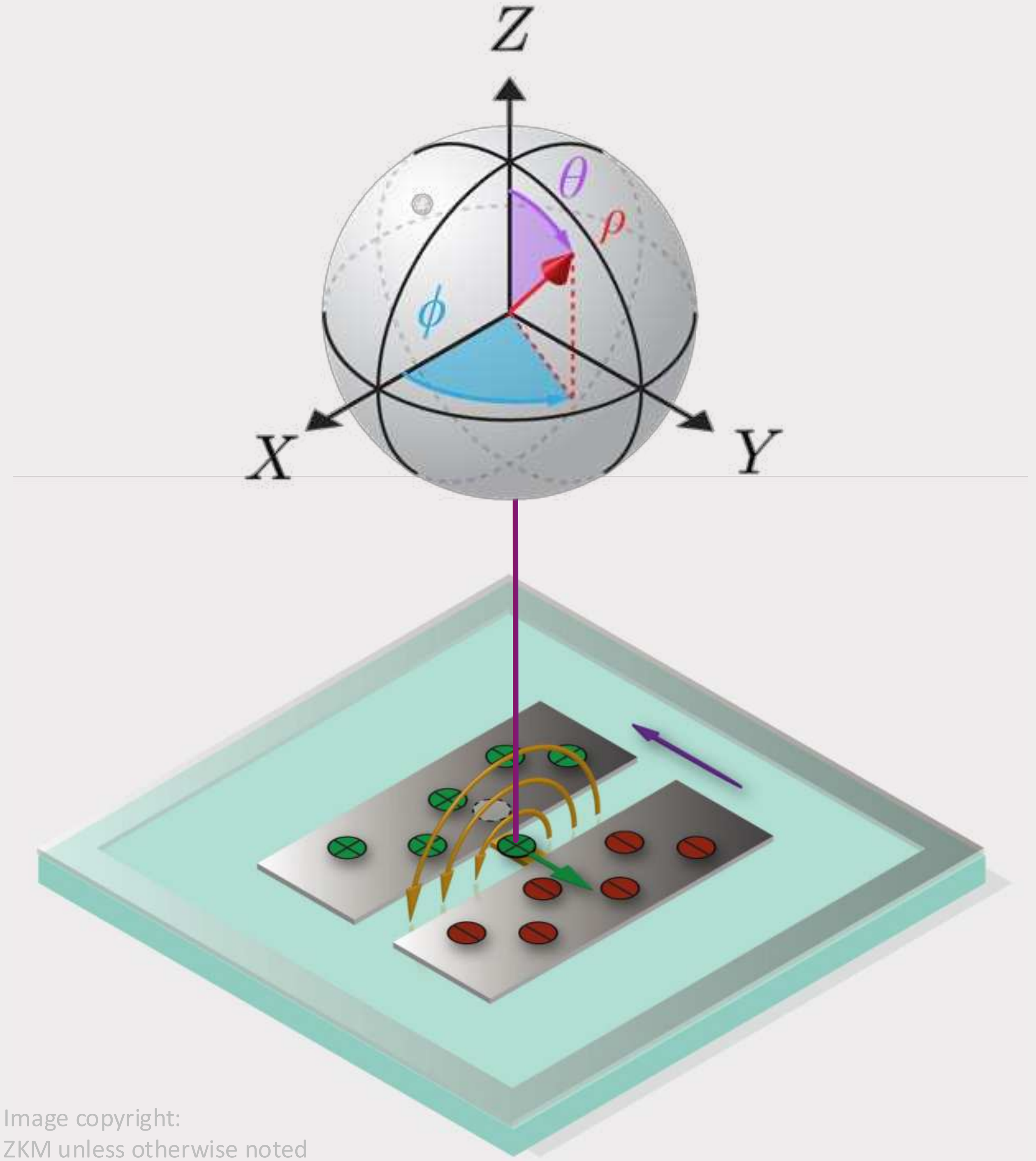
Circuit Quantum Electrodynamics







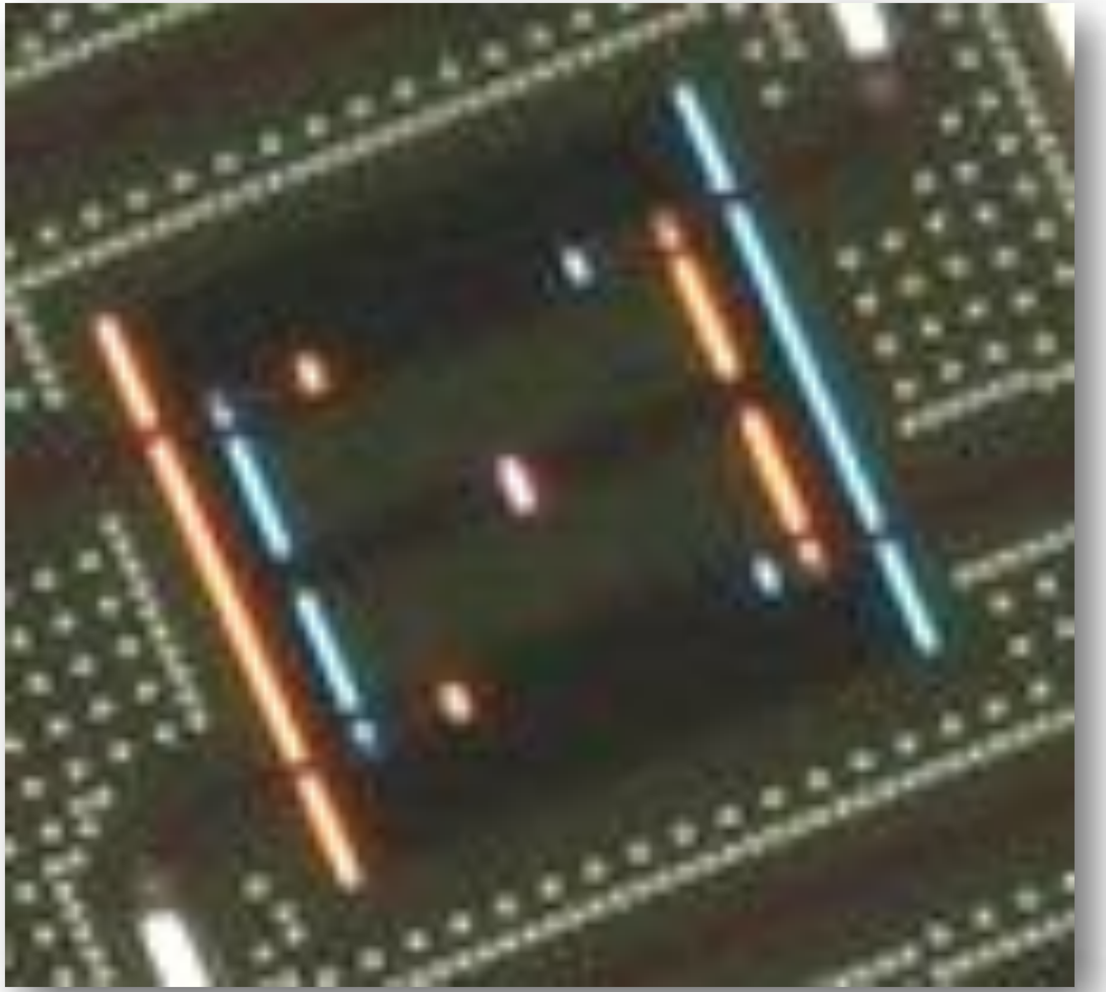


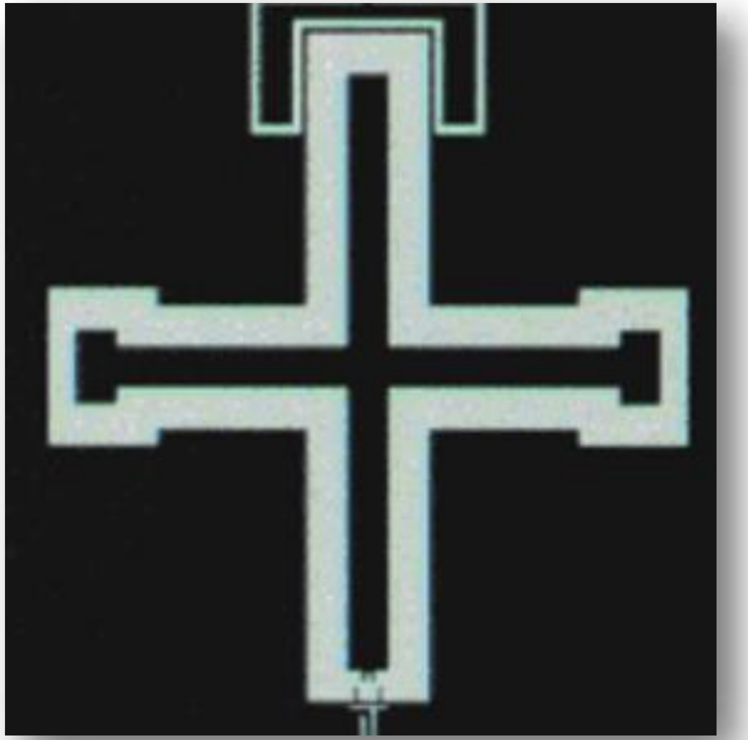
To quantum,

start classical

Classical oscillator

Transmon qubit





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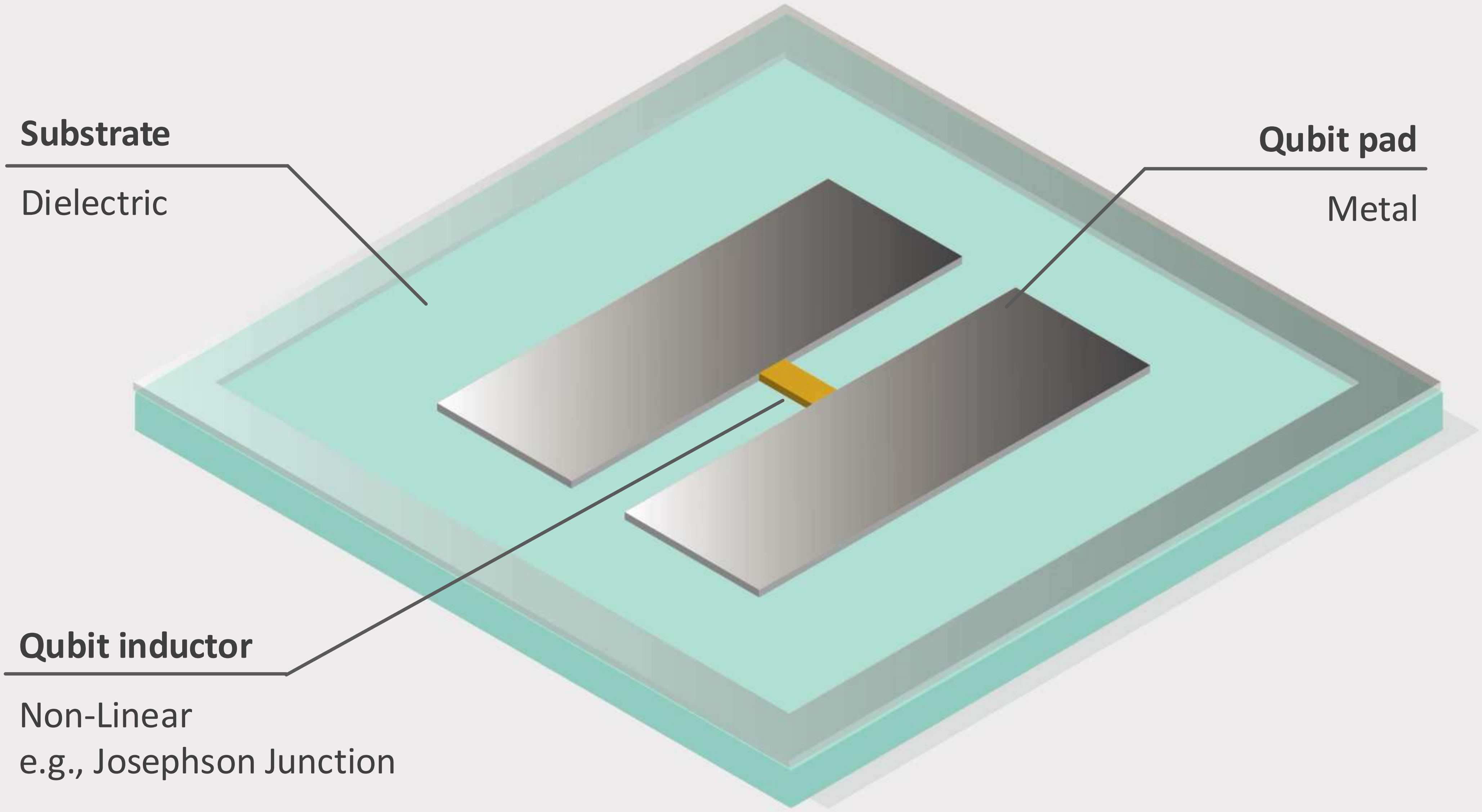
Transmon qubit



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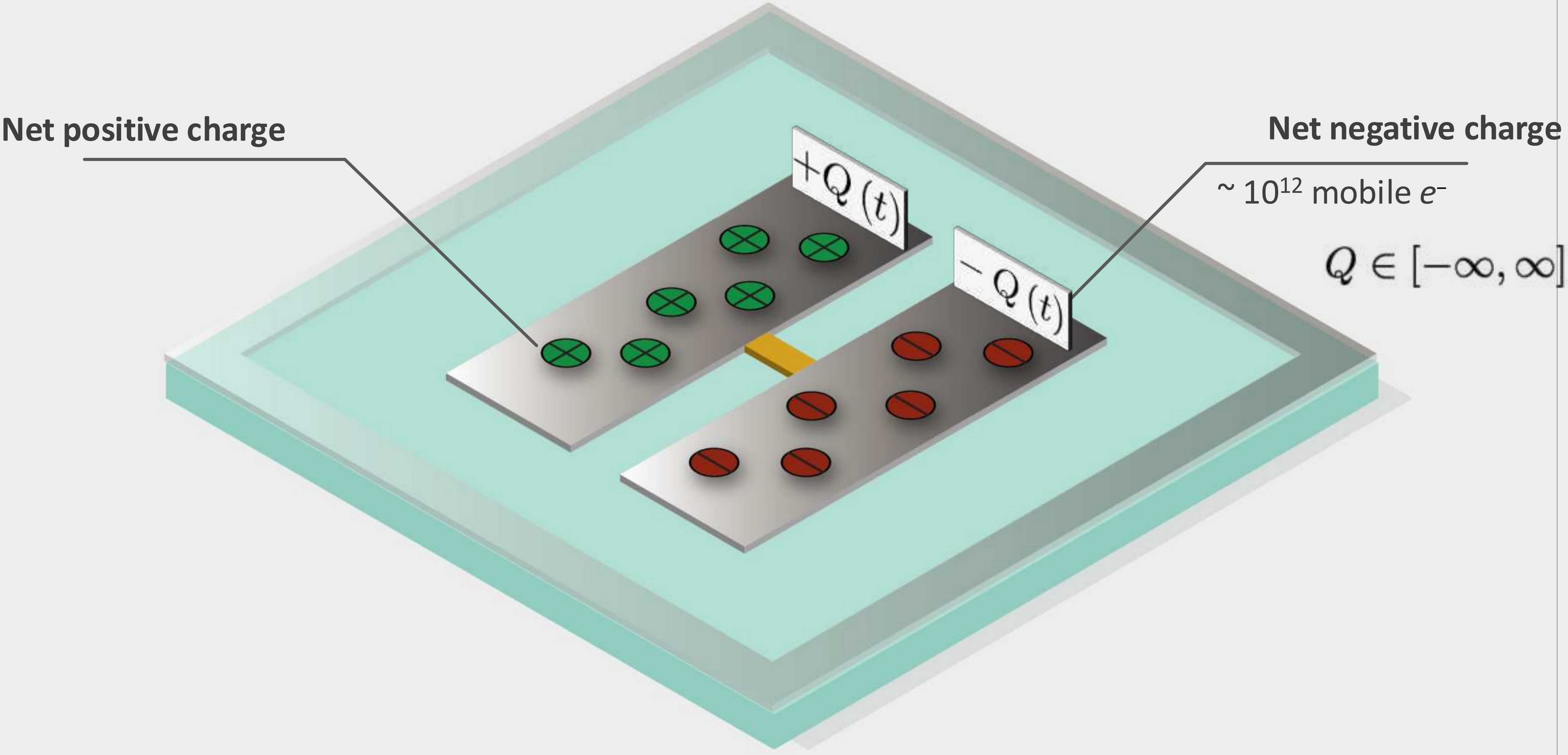
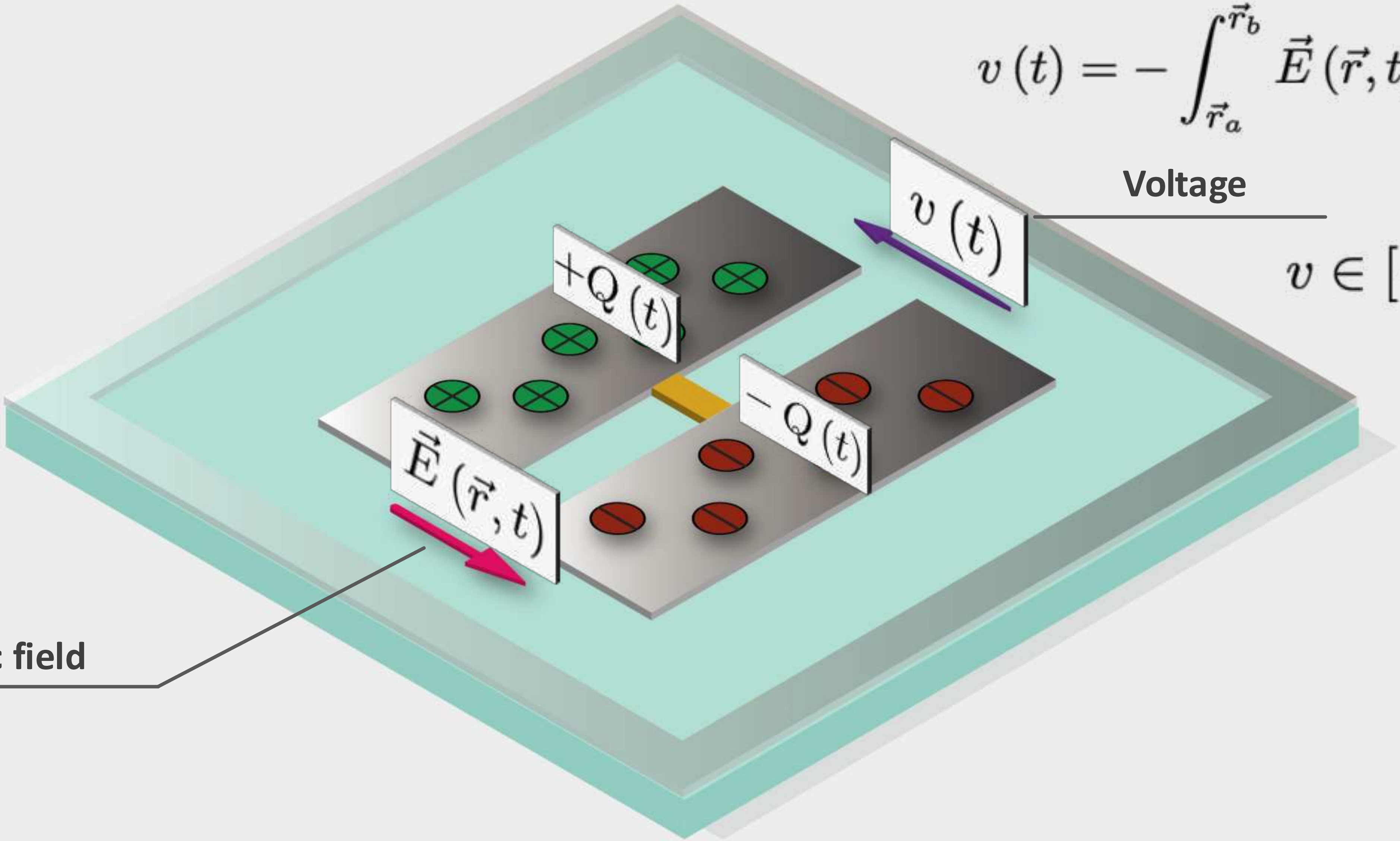
Transmon qubit: charge



Electric field and voltage

$$v(t) = -\int_{\vec{r}_a}^{\vec{r}_b} \vec{E}(\vec{r}, t) \cdot \mathrm{d}\vec{l}(\vec{r})$$



Voltage

$$v \in [-\infty, \infty]$$

Electric field

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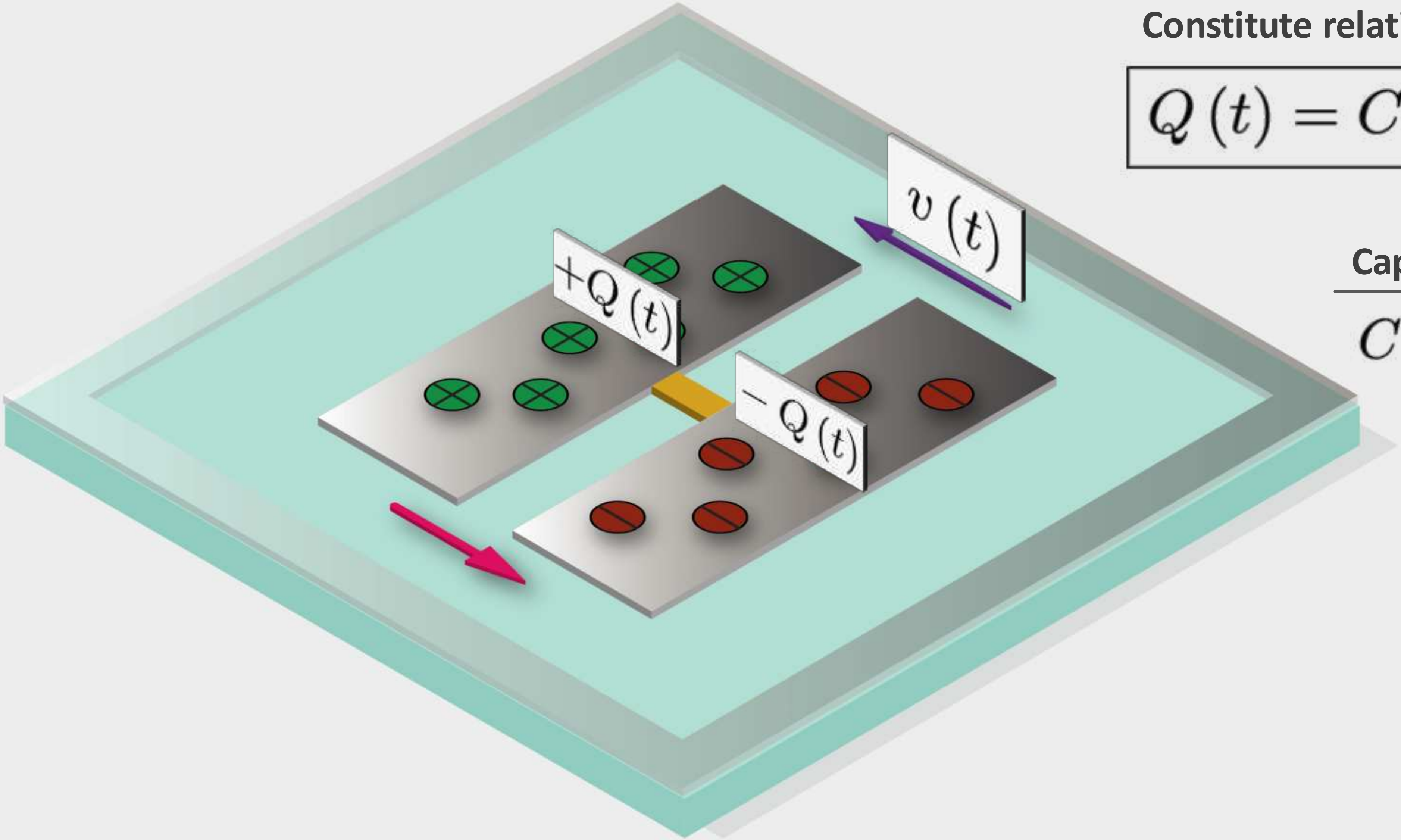
Charge and capacitance



Constitute relationship

$$\boxed{Q(t) = Cv(t)}$$

Capacitance

$$C \in (0, \infty)$$

For a good discussion, see "*The Feynman Lectures on Physics Vol. II Ch. 22: AC Circuits.*" Caltech.

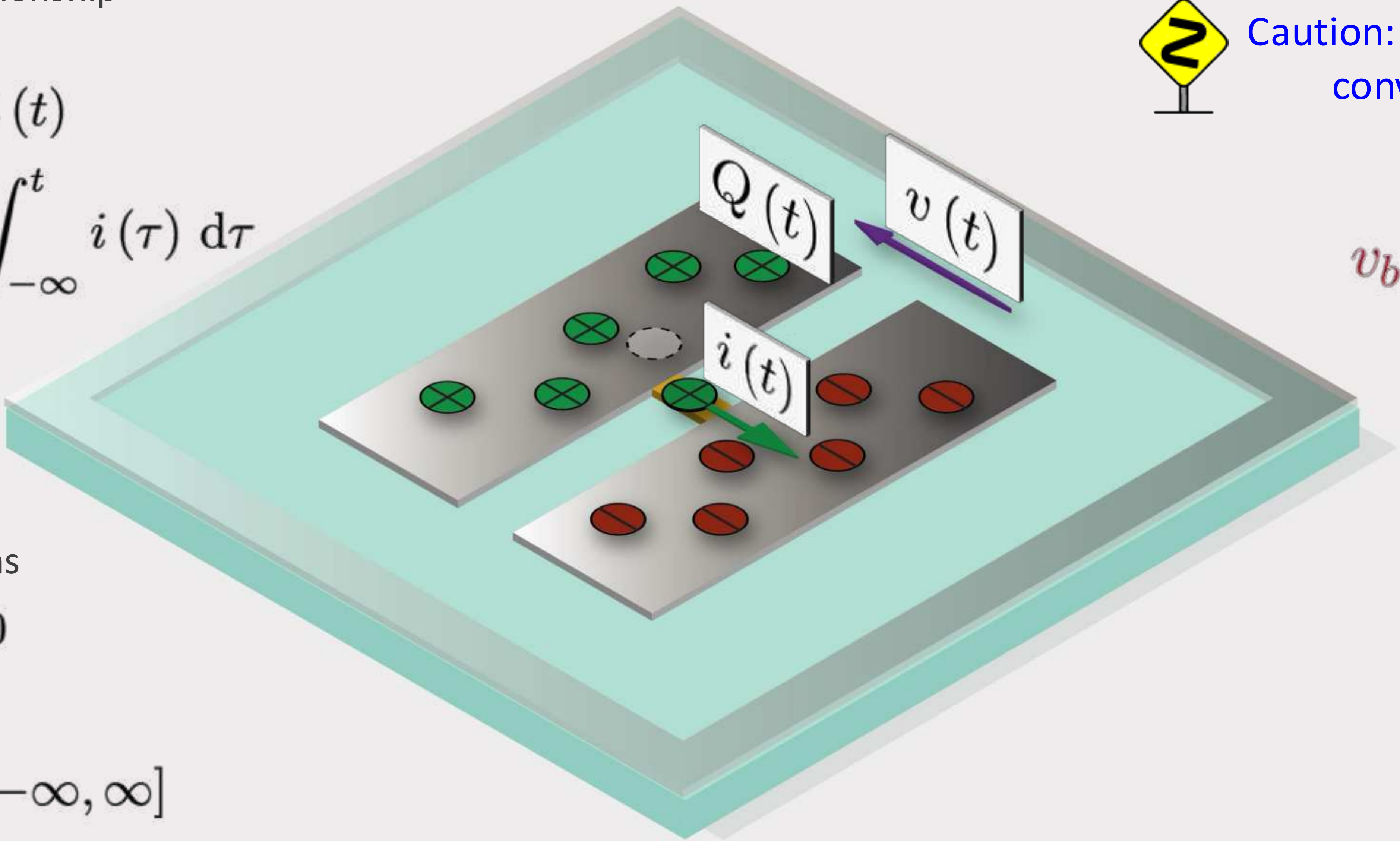
Charge and current

Conservation of charge

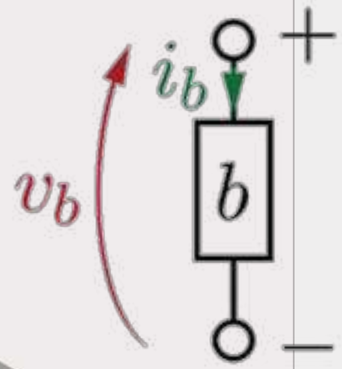
Universal relationship

$$\frac{\mathrm{d}}{\mathrm{d}t}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^{t} i(\tau)\,\mathrm{d}\tau$$



Caution: Passive sign convention



Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$

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Magnetic flux and inductance

Faraday's law of induction

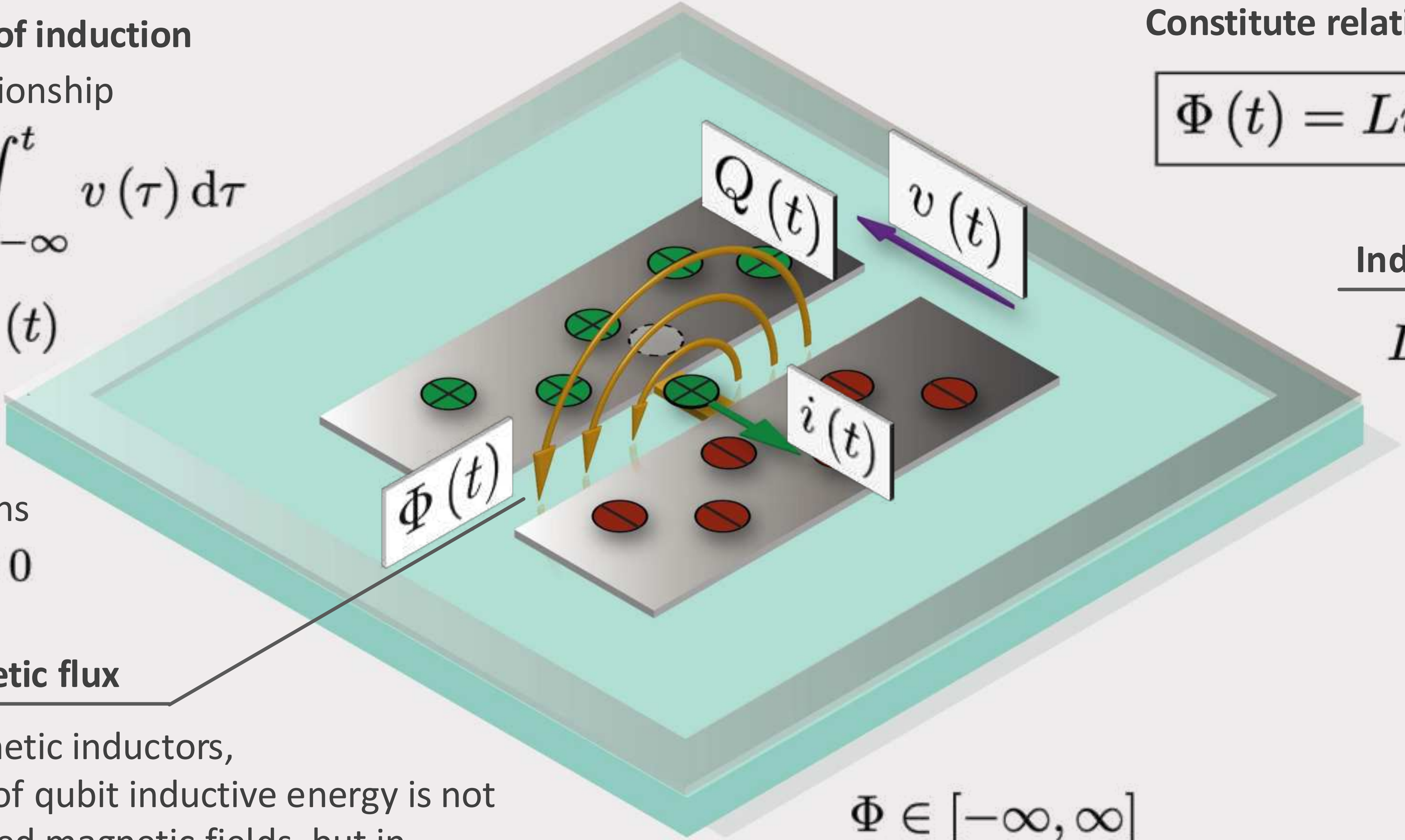
Universal relationship

$$\Phi(t) = \int_{-\infty}^{t} v(\tau)\,\mathrm{d}\tau$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$



Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not in stored magnetic fields, but in kinetic inductance

$$\Phi \in [-\infty, \infty]$$

Constitute relationship

$$\boxed{\Phi(t) = Li(t)}$$

Inductance

$$L \in (0, \infty)$$

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Power and energy

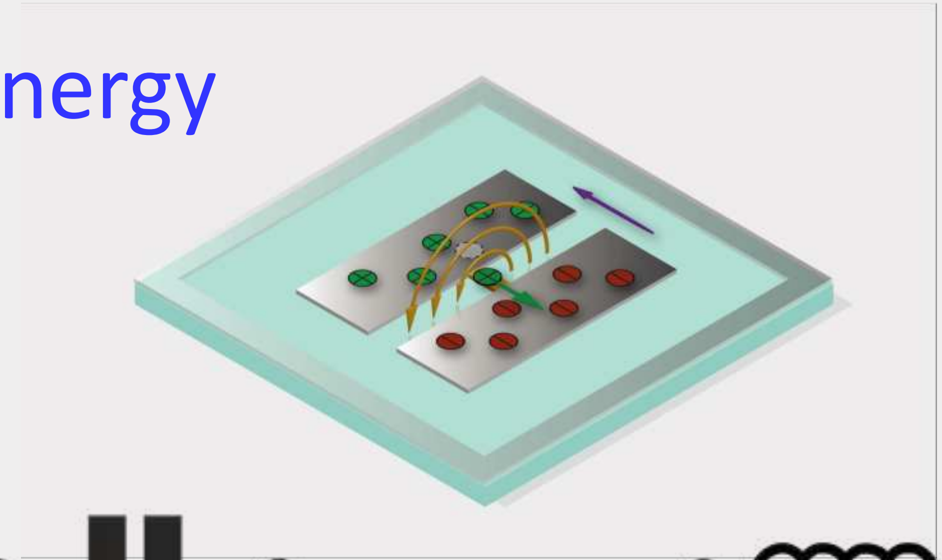
Universal

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}\left(t\right)=p\left(t\right)\equiv v\left(t\right)i\left(t\right)$$

Energy stored *in* (delivered *to*) component

Instantaneous power flowing *to* component

$$\mathcal{E}\left(t\right)=\int_{-\infty}^{t}p\left(\tau\right)\,\mathrm{d}\tau$$



$$\mathcal{E}_{\mathrm{cap}}\left(\dot{\Phi}\right)=\frac{1}{2}C\dot{\Phi}^{2}$$

$$\mathcal{E}_{\mathrm{ind}}\left(\Phi\right)=\frac{\Phi^{2}}{2L}$$

$$\mathcal{E}_{\mathrm{cap}}\left(Q\right)=\frac{Q^{2}}{2C}$$

$$\mathcal{E}_{\mathrm{ind}}\left(\dot{Q}\right)=\frac{1}{2}L\dot{Q}^{2}$$

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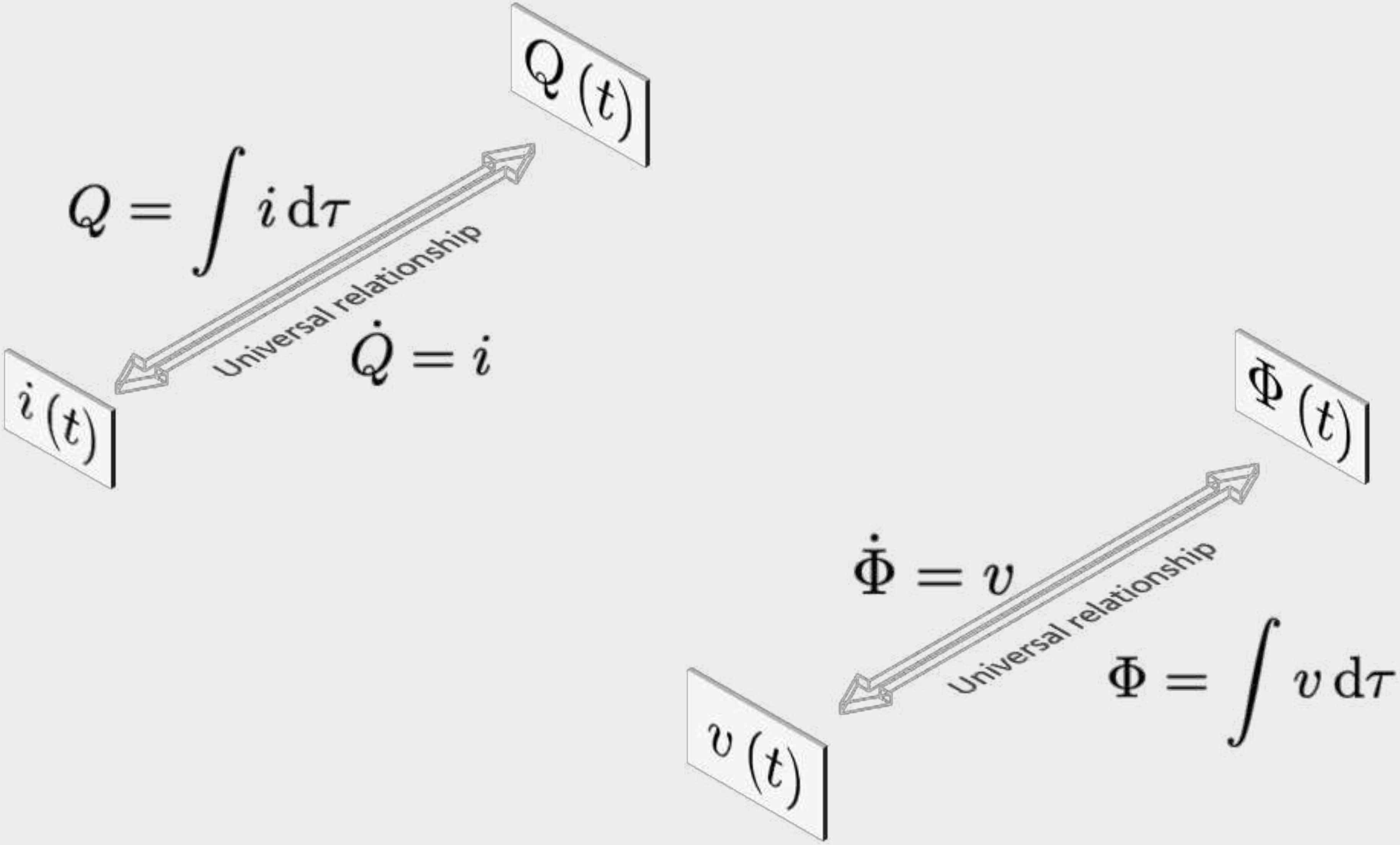
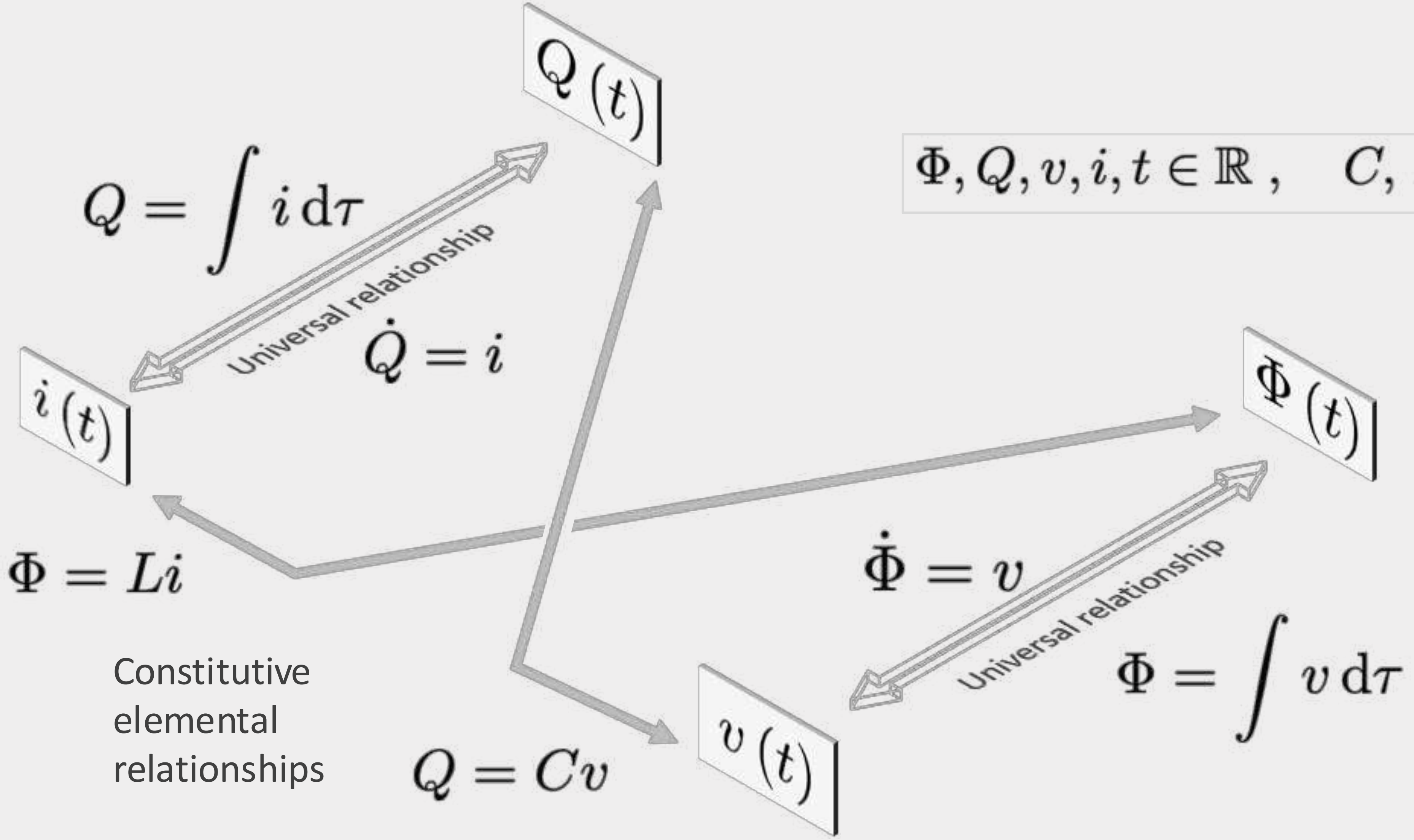
Four fundamental manifestations of electricity

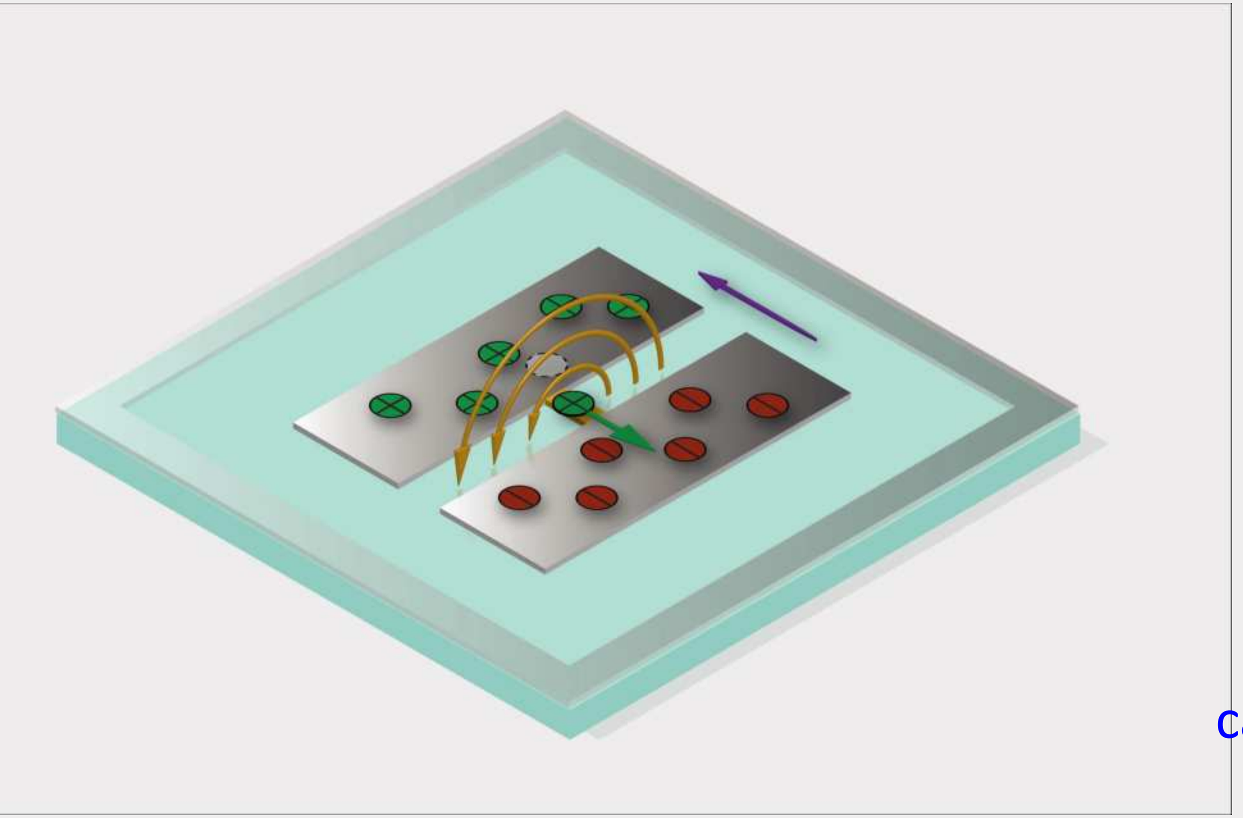


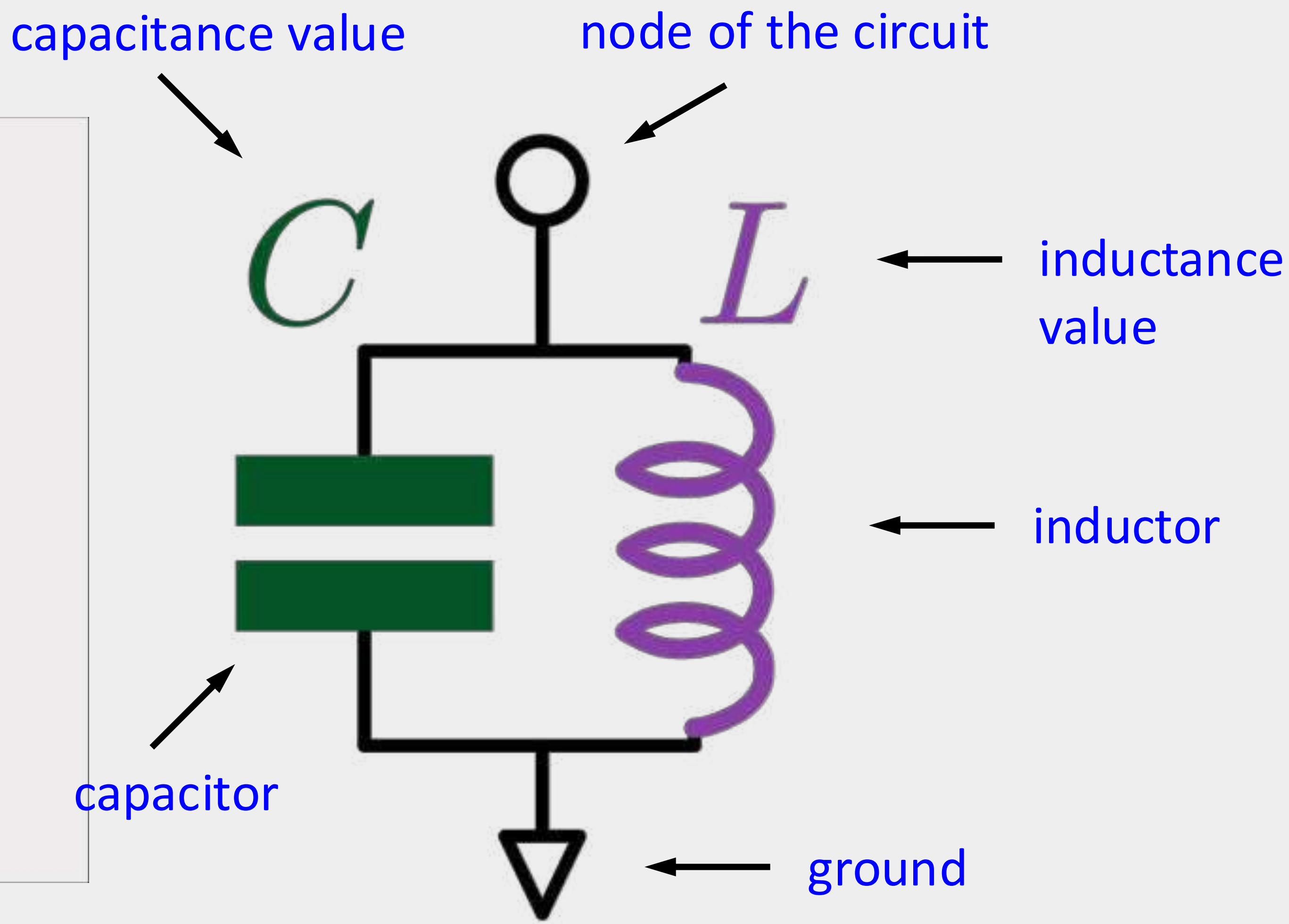
Four fundamental manifestations of electricity

$$\Phi, Q, v, i, t \in \mathbb{R}\,, \quad C, L \in \mathbb{R}_{>0}$$



Electromagnetic oscillator



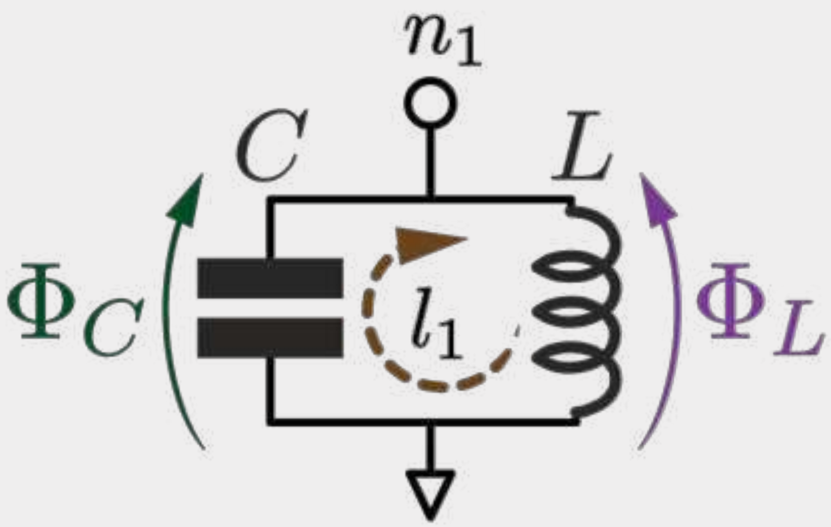


Kirchhoff's network laws*

Conservation of charge

Kirchhoff's current law

$$\sum_{b\in\mathrm{node}} \pm\dot{Q}_b(t)=0$$



Faraday's law of induction

Kirchhoff's voltage law

$$\sum_{b\in\mathrm{loop}} \pm\dot{\Phi}_b(t)=0$$

$$n_1: \quad \dot{Q}_C+\dot{Q}_L=0$$

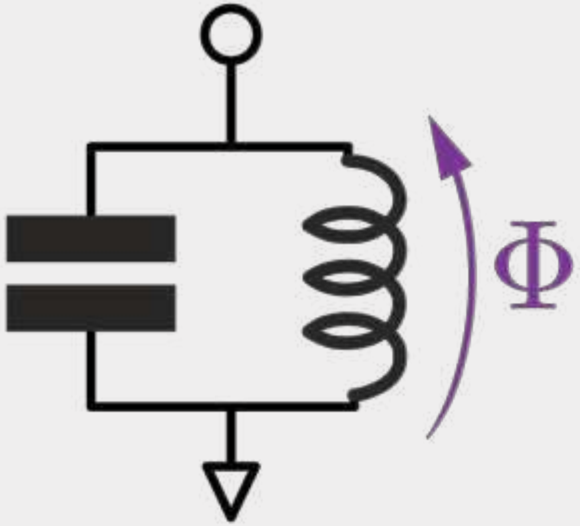
$$C\ddot{\Phi}+L^{-1}\Phi=0$$

$$l_1: \quad \dot{\Phi}_C-\dot{\Phi}_L=0$$

$$\Phi_C=\Phi_L$$

As we will see later, for the Lagrangian description in flux basis, KVL acts as a set of holonomic constraints and KCL as the equations of motion

Oscillator analogy



$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\ddot{\Phi} = -\omega_0^2 \Phi\,, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Resonance frequency

$\dot{\Phi}(t)$ Velocity

$\Phi(t)$ Deviation from equilibrium

Equilibrium position

$x = 0\,, \quad \Phi = 0$

Spring: Svjo

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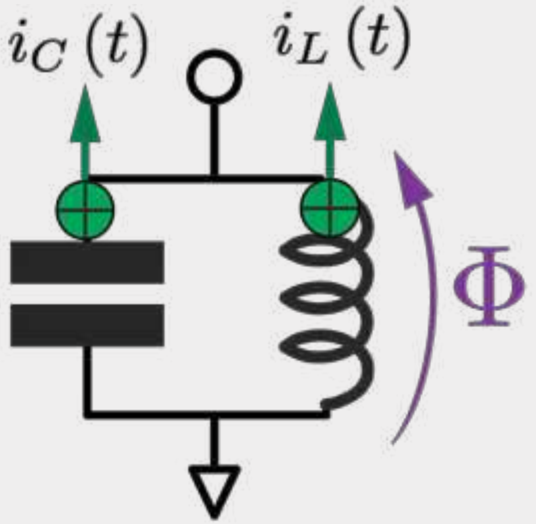
Lagrangian and Hamiltonian*



$$\underbrace{C\ddot{\Phi}}_{i_C(t)} + \underbrace{L^{-1}\Phi}_{i_L(t)} = 0$$

$$\mathcal{L}\left(\Phi, \dot{\Phi}\right) = \mathcal{E}_{\mathrm{cap}}\left(\dot{\Phi}\right) - \mathcal{E}_{\mathrm{ind}}\left(\Phi\right)$$

$$= \frac{1}{2}C\dot{\Phi}^2 - \frac{\Phi^2}{2L}$$

Kinetic energy – Potential energy

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi}$$

Canonically conjugate variable (momentum; charge)

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \frac{\partial \mathcal{L}}{\partial \Phi}$$

Euler-Lagrange equations, $F = ma$

$$\mathcal{H}\left(\Phi, Q\right) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

Kinetic energy + Potential energy

* Temporarily going to assume some minimal knowledge of classical mechanics.
Since we eliminated the KVL constraints, this is a Lagrangian of the second kind.

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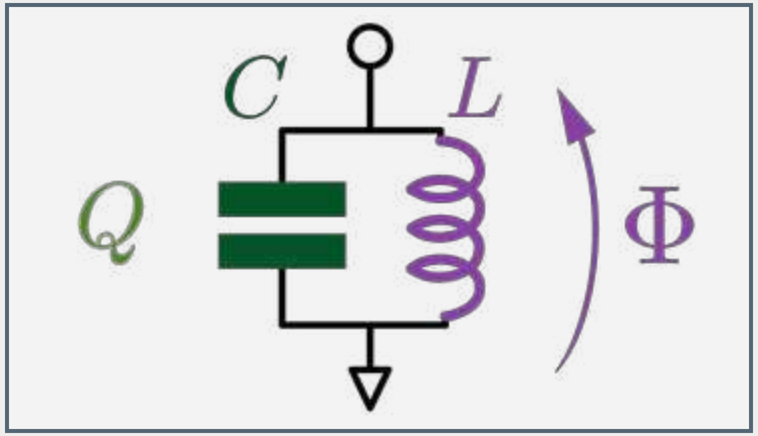
The LC classical harmonic oscillator



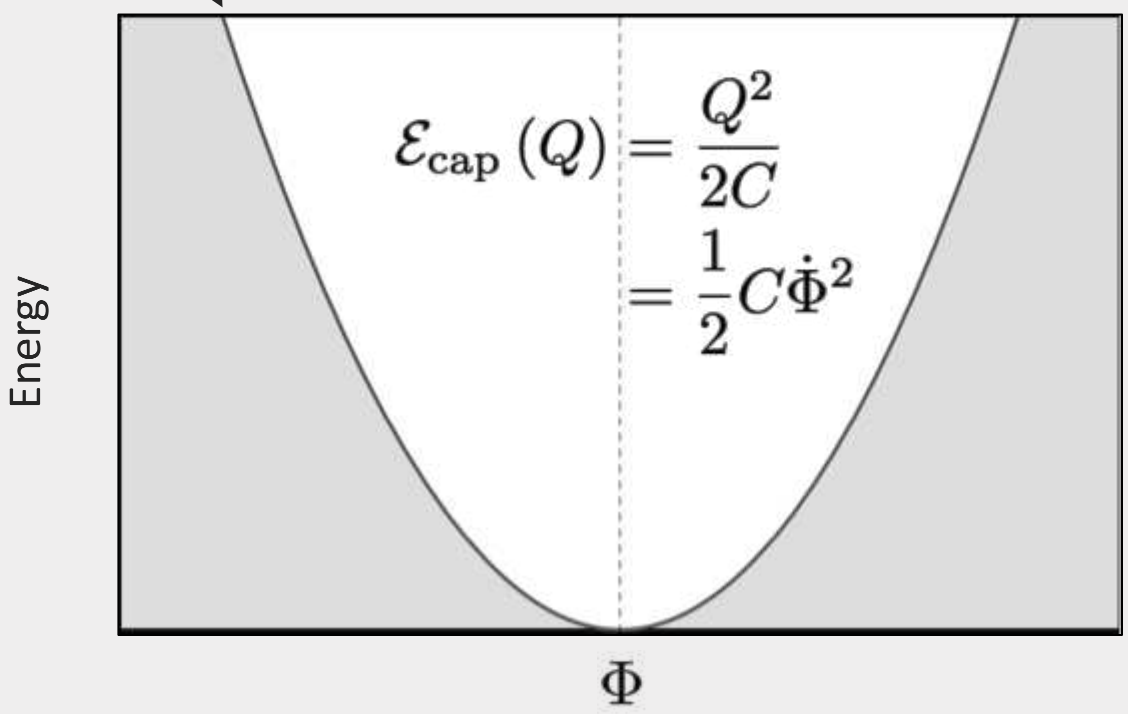
Position: $\Phi \mapsto x$ Momentum: $Q \mapsto p$

Mass: $C \mapsto m$ Spring constant: $L^{-1} \mapsto k$

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \qquad Z_0 = \frac{L}{C}$$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$



Graph: Minev; Spring-mass: Svjo

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“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

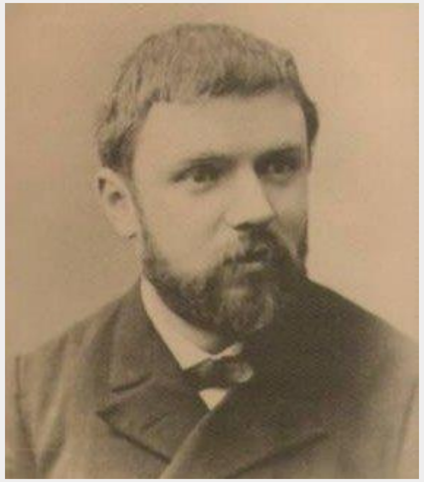


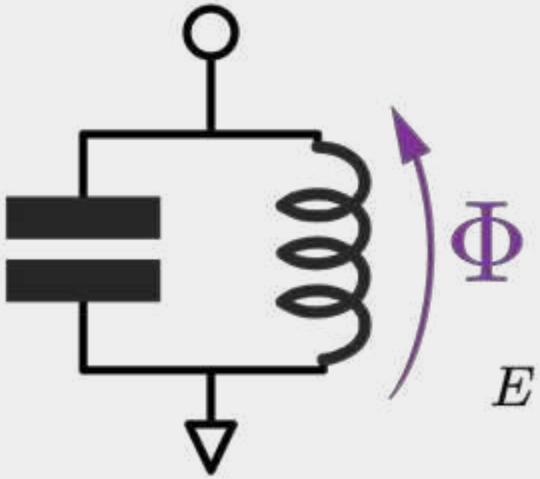
Photo by Eugène Pirou

Hamiltonian dynamics and phase space

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$E = \hbar\omega_0\left(n + \frac{1}{2}\right) = \frac{1}{2}\left(Q^2 + \Phi^2\right)$$

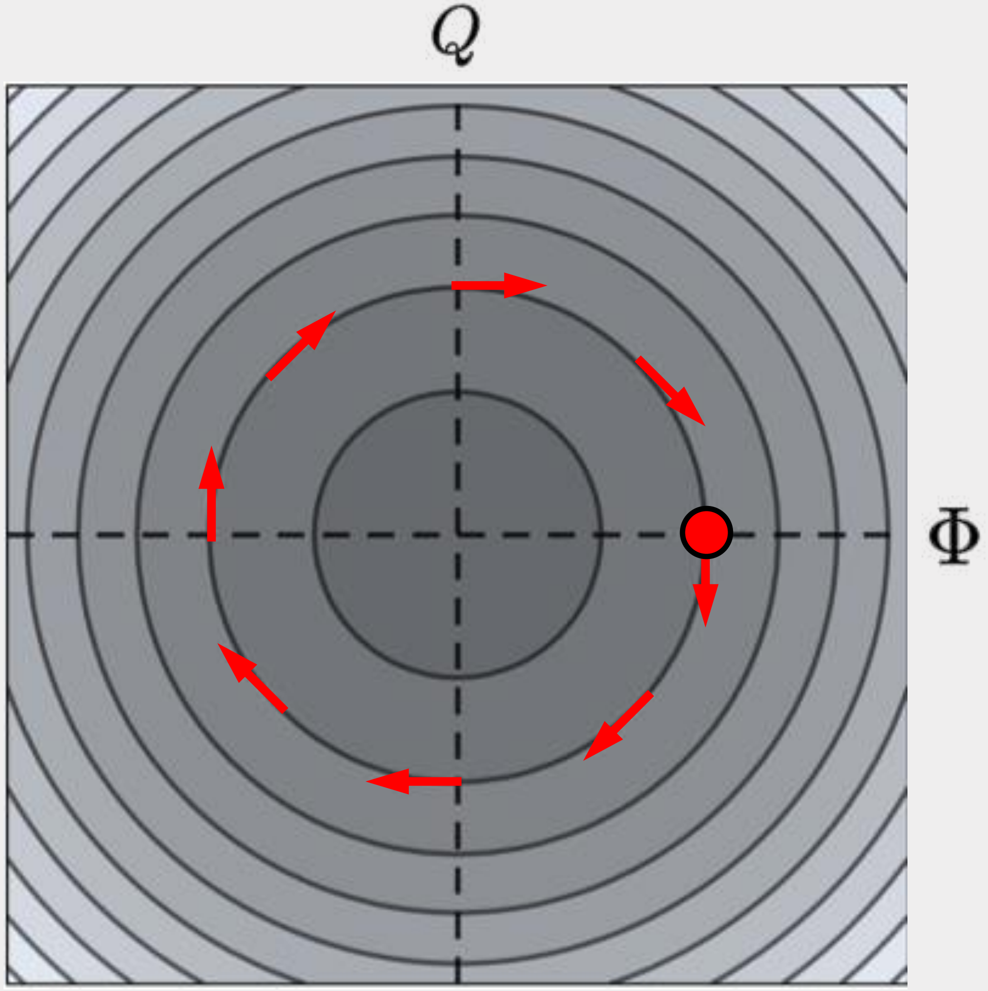
with $L = C = 1$



⋮

$$E = \frac{3\hbar\omega_0}{2}$$

$$E = \frac{\hbar\omega_0}{2}$$



$$\dot{\Phi} = +\frac{\partial \mathcal{H}}{\partial Q} = +Q$$

$$\dot{Q} = -\frac{\partial \mathcal{H}}{\partial \Phi} = -\Phi$$

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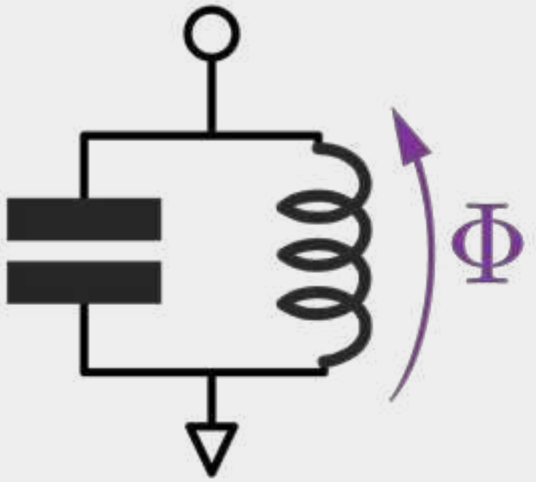
Complex action-angle variable

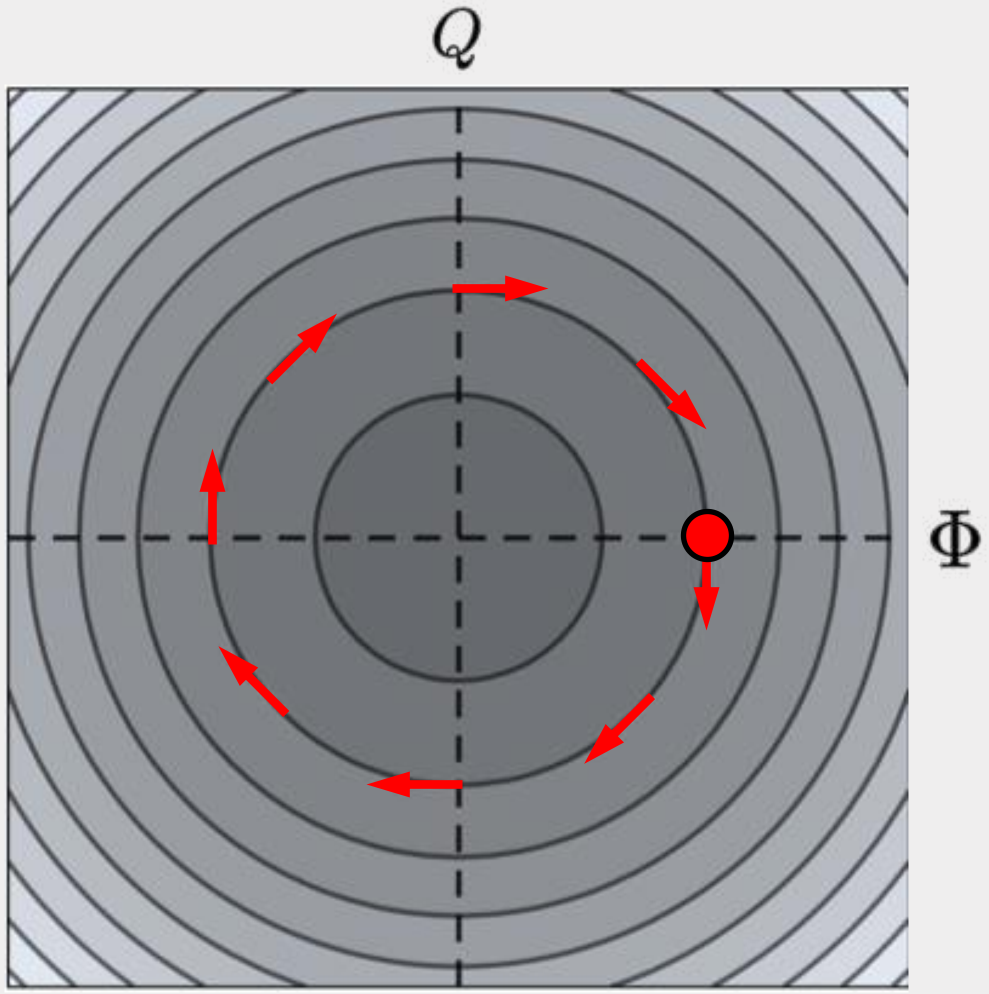
$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\mathcal{H}(\alpha, \alpha^*) = \frac{\hbar\omega_0}{2}(\alpha^*\alpha + \alpha\alpha^*)$$

$$E = \hbar\omega_0 \left(n + \frac{1}{2}\right)$$

$$\alpha(t) = \sqrt{\frac{1}{2\hbar Z}}\left[\Phi(t) + iZQ(t)\right]$$





Classical analog of the bosonic ladder operator

$$\alpha(t) = \alpha(0)\, e^{-i\omega_0 t}$$